Restricted partitions, domination polynomials, and the Kanter factor

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Abstract

Let $p_A(n)$ count partitions of n into parts from a fixed set A of positive integers with greatest common divisor one. We prove that $p_A(h+n)/(p_A(h)p_A(n))$ tends to zero uniformly for $n \geq h \to \infty$ whenever $|A| \geq 2$. If A is infinite, the ratio decays faster than every fixed power of $1/h$. The proof cuts a partition at a chosen sum and bounds the number of partitions that produce the same two lists. A separate finite calculation determines all exceptions in the square-partition inequality.

Random deletion from a dominating set proves ultra-log-concavity for powers of paths and cycles. For trees, removing one leaf can move the most frequent size of a dominating set by an unbounded distance, even when both sequences are unimodal. In probability, a contour estimate and a compact interval certificate give the generalized gamma convolution range $0 < \beta \leq 4/5$ for the Kanter factor K_β. A different prediction for the shifted positive free stable law fails throughout $23/25 \leq \beta < 1$, where its density cannot be a gamma scale mixture of shape $3/2$. A shifted gamma integral establishes the coefficient identity at the limiting stable edge in all orders. Finite bell shape also distinguishes interior derivative zero counts from endpoint conditions. The remaining results concern sharp Jacobi endpoint comparison, global Bessel bounds, and the fractional completely monotonic degrees of two trigamma remainders.

Scope. Page 2 identifies the ten source statements. Six are proved and four are disproved under their printed hypotheses. Established special cases and closely related earlier results are credited where used. The manuscript includes its five finite verifiers. These are mathematical claims submitted for scrutiny. Worldwide priority and comparative rank within the entire source corpus have not been established.

1 Original questions and proof locations

Entries 3 and 4 involve ordinary domination on two different graph families. Entries 5–7 ask about two probability laws and an endpoint expansion in the same source. A single entry is used for each source conjecture with several clauses.

	Source	Conclusion	Page
1	Gajdzica, Conj. 6.5 (2024 preprint)	True for arbitrary allowed parts. The ratio decays superpolynomially if the set of parts is infinite.	4
2	Roy, Conj. 3.2 (2024 preprint)	True. All equalities and failures for partitions into squares agree with the proposed table.	7
3	Galvin–Zhang, Conj. 1.8 (2024)	True. Domination coefficients divided by the corresponding binomial coefficients are log-concave for every power of both paths and cycles.	8
4	S. Zhang, Conj. 4.1 (2021)	False. Removing one leaf can move every mode arbitrarily far while preserving unimodality.	10
5	Hasebe–Simon–Wang, Conj. 1 (2018/2020)	True. The Kanter variable is a generalized gamma convolution exactly for $\beta \leq 4/5$.	12
6	Hasebe–Simon–Wang, Remarks 2(b), 10(b)	False. The shifted positive free stable law has the proposed gamma mixture for $\beta \leq 4/5$ and fails it throughout $23/25 \leq \beta < 1$.	14
7	Hasebe–Simon–Wang, Remark 12	True. The stable-edge coefficient equals the proposed partial sum of Stirling coefficients in every order.	15
8	Booker–Strömbergsson–Then, Conj. 1 (2013)	True in every parameter-derivative order. A two-weight extension gives a necessary and sufficient parameter range.	16
9	Baricz–Wu, concluding remark 6 (2009)	False. Each proposed global extension fails. We determine the exact order range for the first two inequalities.	17
10	Qi, Remark 7.4 (preprint 2013, journal 2020)	False. The exact replacement degrees are 2, 3, or a uniquely specified value strictly between them.	18

From auxiliary problems to the original questions. Three transfers are especially direct.

1. For partitions, ask how many input partitions can produce the same pair of lists after a cut. A fixed finite subset of the allowed parts gives many ways to fill both lists back to prescribed sums. A second cut deals with the inputs that leave too little room on the first pass. This gives a bound uniform in the larger input, including for sets having arbitrarily large gaps.

2. For graph powers, ask whether adding a bounded part to a composition of fixed total makes the sum of two distinguished parts stochastically smaller. The comparison translates to the probability that deleting a random member of a dominating set preserves domination. That probability gives the ratio of successive normalized coefficients.

3. For the Kanter factor, ask when the argument of a boundary value of its transform is increasing. The known probabilistic criterion turns this into a two-parameter sign problem. A contour with a uniformly controlled oscillatory integral deals with large arguments. On the compact remainder, derivative bounds turn finitely many rounded calculations into estimates on the entire parameter continuum.

Connection with the source corpus. We consulted the Verifiable Mathematical Reasoning Corpus, revision `ecd87de3348dc`. Item [076] uses a first crossing of a prefix deficit to reconstruct combinatorial objects. Here the crossing of a sorted sum is used for an estimate on the number of preimages. Items [067] and [083] provided two other points of departure: converting distributional questions to signs of transforms, and separating compact inequalities from bounds at the boundary of an unbounded domain. The statements about the Kanter factor and the curvature of the beta manifold are different. The theorems in this manuscript use the mathematical inputs specified in their respective proofs. Earlier results of the corpus have not been counted as resolutions in the table.

Related graph problem and priority. The existence of ordinary domination polynomials with multiple peaks had already been reported in 2026, including S. J. Silverman's July preprint and L. R. Chalmers's October formalization archive. The ordinary-domination conjecture is excluded from the table. Entry 3 concerns powers of two specified graph families and a stronger inequality. Entry 4 compares modes for pairs of unimodal trees. Boyer et al.'s zero-forcing conjecture also has a prior counterexample: P. Graham, *A graph whose zero-forcing polynomial is not unimodal*, 1 November 2026. Appendix A gives a smaller graph and an arbitrary-number-of-peaks construction. This is outside the ten.

Verification. The source contains complete listings for the square-partition finite check, the independent zero-forcing enumeration, and the interval checks for the Kanter and trigamma results. A fifth verifier uses rational polynomials for the shifted-law obstruction. The finite-bell-shape results in the appendices distinguish exact printed zero counts from the older endpoint convention. Under that convention relevant positive implications were already proved by Kwaśnicki and Simon. They are excluded from the ten. Appendix D gives a further application of a fractional-kernel comparison from the previous project.

2 Submultiplicativity for partitions with arbitrary allowed parts

For $A \subseteq \mathbb{N}$ let $P_A(j)$ count unordered partitions of j with all parts in A. Distinct partitions record only multiplicities. Conjecture 6.5 in the 2024 preprint of K. Gajdzica, *On the Bessenrodt–Ono type inequality for a wide class of A-partition functions*, arXiv:2401.16267, published in Ramanujan J. 66 (2025), article 80, asks if $P_A(h)P_A(n) > P_A(h+n)$ for all sufficiently large h, n, for every finite or infinite A with $|A| \geq 2$ and $\gcd A = 1$. There is no regularity assumption on the spacing of the parts.

Theorem 2.1. *Under exactly these hypotheses,*

$$\lim_{H\to\infty} \sup_{n\geq h\geq H} \frac{P_A(h+n)}{P_A(h)P_A(n)} = 0.$$

Furthermore the convergence is uniform over all A containing a prescribed finite set C with $\gcd C = 1$ and $|C| \geq 5$.

Proof. For a fixed finite set C of $r \geq 2$ parts and greatest common divisor one, $p(j) = P_C(j)$ has the estimate

$$p(j) = \frac{j^{r-1}}{(r-1)!\prod_{a\in C} a} + O_C((1+j)^{r-2}). \tag{A1}$$

In fact $\prod_{a\in C}(1-z^a)^{-1}$ has its unique pole of order r at 1, with leading coefficient $1/\prod a$ in powers of $(1-z)^{-1}$. All other poles are roots of unity of order at most $r-1$ as poles, so partial fractions give (A1). For finite A this already proves the result, since $n+h \leq 2n$. For infinite A take a finite gcd-one subset $C \subset A$ and augment it until $r = |C| \geq 5$. Such a subset exists by stabilization of the gcd of the initial segments of A. We prove the stronger uniform claim with this C held fixed.

Counting the originals of a cut. Let $D = A \setminus C$. We will assign a partition of $n+h$ to a pair (L, R) of D-partitions, called its prefix and suffix, leaving at most three individual D-parts unassigned. Define $u = h - |L|$, $v = n - |R|$, where bars denote sums of parts. All pairs of full partitions of (h, n) with precisely these two D-parts multisets are counted by $p(u)p(v)$. As (L, R) varies these classes are disjoint and their sizes sum to $P_A(h)P_A(n)$.

Suppose first that the original C-parts have total weight $s \leq 2\min(u, v)$ and no more than three D-parts were removed. For fixed (L, R) and s the missing parts sum to $u + v - s$. There are $O((1+u+v)^2)$ possibilities, by counting ordered compositions into at most three positive terms and allowing an empty composition. The original D-multiset is the union of the known multisets and the missing parts. There are $p(s)$ choices of its C-multiset. Write $x = \min(u, v)$, $y = \max(u, v)$. By (A1) the number of possible originals in this class is at most $K_C(1+x)^r(1+y)^2$. For x, y sufficiently large the ratio to the class size $p(u)p(v)$ is therefore at most

$$K'_C x\, y^{3-r} \leq K'_C x^{4-r}. \tag{A2}$$

In a second type of assignment no part is deleted and all D-parts go on the right, with $s \geq 2h$. Here $u = h$, $v = s - h \geq h$ and the number of possible originals is bounded by $p(s) = p(h+v)$. For sufficiently large h the ratio to $p(h)p(v)$ is at most $K''_C h^{1-r}$, again by (A1). Both estimates allow arbitrary D.

A cut preserving space for the finite set. Fix $\varepsilon > 0$ and choose an integer B so large that (A2) is at most $\varepsilon/4$ whenever $x \geq B$. Take $h \geq 4B$ sufficiently large that the second type of assignment also costs at most $\varepsilon/4$ of a class. Each partition with $s \geq 2h$ receives precisely this second assignment.

For $s < 2h$ order all D-parts nondecreasingly. Their total $h + n - s$ exceeds $h - s/2 > 0$. Remove the first part q for which the cumulative sum reaches $h - s/2$. If L_0, R_0 are the preceding and following lists, respectively, set $u_0 = h - |L_0|$, $v_0 = n - |R_0|$. The definition of the crossing gives

$$s/2 < u_0 \le s/2 + q, \qquad v_0 = s + q - u_0 \ge s/2. \tag{A3}$$

If $u_0 < B$, remove the largest remaining part of L_0. If $v_0 < B$, remove the smallest remaining part of R_0. Each list from which a removal is required is nonempty: an empty L_0 would give $u_0 = h \ge B$, and an empty R_0 would give $v_0 = n \ge B$. After these operations at most three parts have been removed. If the new capacities u, v are both at least B, use the new prefix and suffix as (L, R). Inequality (A3) ensures $s \le 2\min(u, v)$, so the first estimate applies. Thus the total number of partitions assigned so far is at most $(\varepsilon/2)P_A(h)P_A(n)$, by summing the two estimates over all classes.

The partitions left over. We show that every unassigned partition has at least $h - 2B$ of its weight in $C' = C \cup (A \cap [1, B])$. If the final left capacity is below B, the largest part of L_0 was removed and was less than B. Thus the entire list L_0 consists of parts less than B, of total $h - u_0 > h - B$. If instead the final right capacity is below B, the smallest part of R_0 was less than B, so $q < B$ by the ordering. Also $s/2 \le v_0 < B$. Now $u_0 \le s/2 + q < 2B$ by (A3), and every part of L_0 is at most $q < B$. In this case $|L_0| > h - 2B$, as required.

Apply the construction again to these original partitions, with their original parts intact and the new finite set C' in place of C. Write s' for its weight in a partition. If $s' \ge 2h$, use the assignment with every part outside C' on the right. If $s' < 2h$, make the sorted cut and remove only its crossing part. By (A3) both capacities are at least $s'/2 \ge (h - 2B)/2$. Consequently the total number of partitions assigned in this second pass is at most $(\varepsilon/2)P_A(h)P_A(n)$ for all sufficiently large h, by the two preceding estimates for C'. There are only finitely many possibilities for C' once C, B are fixed, so the threshold can be taken independent of the ambient set A. Adding the two counts proves the theorem. In particular the strict inequality in Conjecture 6.5 follows by taking $\varepsilon < 1$. □

Gajdzica's original finite reduction (Theorem 1.4) imposes separation on the allowed parts and verifies several special sets. The sorted split of an individual partition has precedents in combinatorial proofs of Bessenrodt–Ono inequalities, including that source. Here the number of preimages is compared with the number of ways to fill both sums from a finite set. The second cut deals with every failure of the first by using only bounded part sizes. This supplies the arbitrary-set statement and the uniform vanishing ratio.

Corollary 2.2. *Suppose the weight j has $c(j)$ distinguishable colors, with $0 \le c(j) \le M(1 + j)^\beta$ for fixed $M, \beta \ge 0$. Allow each colored part arbitrary multiplicity. If the occupied weights have gcd one and there are at least two colored part types, the same vanishing ratio holds for the colored partition counts.*

Proof. For finitely many colored part types the finite-product proof of (A1) applies with multiplicities. Otherwise choose a gcd-one finite set of $r > 4 + 3\beta$ colored part types. Order all remaining types by weight with a deterministic order for ties. Each deleted part has at most $M(1 + u + v)^\beta$ colors. With at most three deletions, the upper bound preceding (A2) gains a factor $O_M((1 + y)^{3\beta})$. Its ratio to class size is $O_{C,M}(xy^{3+3\beta-r}) = O_{C,M}(x^{4+3\beta-r})$, tending to zero. There are finitely many colored types up to B, so the second cut and the rest of the argument apply unchanged. □

Corollary 2.3. *For every finite gcd-one set C of $r \ge 4$ distinct part sizes, uniformly in $A \supseteq C$ and $n \ge h \to \infty$,*

$$\frac{P_A(h + n)}{P_A(h)P_A(n)} = O_C\big(h^{-(r-3)/2}\big).$$

Consequently, for each infinite A of gcd one and every $K > 0$ the quotient is $O_{A,K}(h^{-K})$ uniformly in $n \geq h$. For a finite gcd-one set A of $r \geq 2$ part sizes the corresponding supremum over $n \geq h$ is of order h^{1-r}.

Proof. In the first cut use $B = \lfloor \sqrt{h} \rfloor$ and put aside every case where both $u_0, v_0 < B$. In such a case $q = u_0 + v_0 - s < 2B$ and $|L_0| > h - B$. The successful cuts now have at most two deleted parts. The upper count preceding (A2) drops to $O_C((1+x)^r(1+y))$ because two missing weights of prescribed sum have only $O(1+y)$ possibilities. Dividing by $p(u)p(v)$ gives a cost $O_C(xy^{2-r}) = O_C(B^{3-r})$. The first-pass no-deletion case still costs $O_C(h^{1-r})$.

For every partition put aside, the parts in L_0 have weight greater than $h - 2B$ and each has size at most $2B$. This was proved above for an unsuccessful single-side removal, and it holds by construction when both sides are small. One size $d \leq 2B$ in D must therefore carry at least $(h-2B)/(2B) \geq h/(4B)$ of the original weight for large h. Choose one such size by a fixed tie-breaking rule. Use $C_d = C \cup \{d\}$ alone for this partition's second cut. The counts $p_d(t) = P_{C_d}(t)$ satisfy

$$p_d(t) \asymp_C t^{r-1}(1 + t/d) \qquad (t \geq t_0(C)),$$

with constants independent of d. Sum $p(t - ld)$ for $0 \leq l \leq t/d$, using $0 \leq l \leq t/(2d)$ for the lower bound. The analogous upper bound with $1 + t$ holds for all $t \geq 0$.

If the new core weight is at least $2h$, the no-deletion estimate remains $O_C(h^{1-r})$ for each d, since $(1 + (h+v)/d) \leq (1 + h/d)(1 + v/d)$. Otherwise the second cut deletes only its crossing, so for fixed s, L, R the deleted weight is uniquely determined. Both remaining capacities are at least $h/(8B)$. The class count is now at most $O_C((1+x)^r(1+x/d))$, by summing $p_d(s)$ for $s \leq 2x$. Division by $p_d(u)p_d(v)$ costs $O_C(xy^{1-r}) = O_C(x^{2-r})$. There are at most $2B$ choices of d, giving the combined cost

$$O_C(B^{3-r} + Bh^{1-r} + B(h/B)^{2-r}) = O_C(h^{-(r-3)/2}).$$

For infinite A choose r arbitrarily large. For finite A, (A1) gives the uniform upper bound, and $n = h$ gives the matching lower bound. □

A sharp boundary for universal color bounds. Let $W : \mathbb{N} \to [1, \infty)$ be a prescribed upper bound on the number of colors at each weight. Every coloring $c(j) \leq W(j)$ with gcd-one occupied weights and at least two colored part types obeys eventual strict submultiplicativity if and only if W is bounded by a polynomial. Sufficiency was proved above. For necessity start with one color at each of the weights $1, 2$. Having chosen finitely many additional colored weights, let R be the current total number of types. Their partition function $p_0(t)$ is $O((1+t)^{R-1})$ by (A1). If W has no polynomial upper bound, there are arbitrarily large N at which

$$\lfloor W(N) \rfloor > j\, p_0(\lfloor N/2 \rfloor) p_0(\lceil N/2 \rceil)$$

at step j. Choose such an N greater than twice the largest previously selected weight, and introduce $\lfloor W(N) \rfloor$ colors at N. For the resulting infinite colored partition function the denominator at the two halves is unchanged forever, whereas the numerator at N counts at least all the new single-part partitions. The ratio at these balanced pairs therefore tends to infinity along the selected weights. No extra named conjecture is counted for this consequence.

Proposition 2.4. *Three distinct prescribed parts of gcd one suffice for convergence to zero uniformly over all sets containing them, with bound $O_C(h^{-1/5})$ for $n \geq h$. For every prescribed two-element gcd-one set, eventual strict submultiplicativity fails to have a threshold uniform over such supersets.*

Proof. For a three-part core modify the quantitative proof with $B = \lfloor h^{1/5} \rfloor$, $J = \lfloor h^{2/5} \rfloor$. A successful cut with only its crossing deleted costs $O_C(xy^{1-r}) = O_C(B^{-1})$ at $r = 3$. A two-deletion

cut must have had one initial capacity below B, whence $s < 2B$. If its final $y = \max(u, v) > J$, its preimage count is $O_C(B^3 y)$, compared with $p(u)p(v) \gg_C x^2 y^2$; the relative cost is $O_C(B/J)$. If instead $y \le J$, we have $q \le u_0 + v_0 \le 2J$. All of L_0 has parts at most $2J$ and weight $h - u_0 \ge h - J$. Put this case aside. The other unsuccessful cuts still have at least $h - 2B$ of weight on parts in L_0 of sizes at most $2B$. For all partitions put aside, some size $d \le 2J$ outside C therefore carries at least $h/(4J)$ of weight for large h. The single-size second pass then costs $O_C(J(h/J)^{-1} + Jh^{-2})$. The total cost is $O_C(B^{-1} + B/J + J^2/h) = O_C(h^{-1/5})$.

For the opposite direction fix $C = \{a, b\}$, $a < b$, with $\gcd(a, b) = 1$, and for each large h set $A_h = C \cup \{h + 1, \dots, 2h\}$. With $p = P_C$,

$$P_{A_h}(h) = p(h), \qquad P_{A_h}(2h) = p(2h) + \sum_{j=0}^{h-1} p(j).$$

Since $p(j) = j/(ab) + O_C(1)$, the quotient at (h, h) tends to $ab/2$, exceeding one unless $C = \{1, 2\}$. For this last set use $p(j) = \lfloor j/2 \rfloor + 1$. At $h = 2m$ the difference $P_{A_h}(2h) - P_{A_h}(h)^2$ is exactly $m > 0$. □

3 The exception table for multiplicativity of square partitions

Write $P_A(n)$ for the number of unordered partitions of n with parts in a set A of positive integers, with $P_A(0) = 1$ and $P_A(n) = 0$ for $n < 0$. For $A = \{1, 4, 9, \dots\}$ write $P(n) = P_A(n)$. In *Log-concavity and the multiplicative properties of restricted partition functions*, Ann. Comb. (2026), Conjecture 3.2, Arindam Roy gives an exact list of the equalities and failures of $P(h)P(n) \ge P(n + h)$ for $n \ge h \ge 1$. The preprint dates to 2024 (arXiv:2404.03153). Gajdzica, *On the Bessenrodt–Ono type inequality for a wide class of A-partition functions*, Theorem 5.2 (arXiv:2401.16267), had already proved strict inequality for $h, n \ge 12$. This leaves infinitely many comparisons with one argument below 12.

An auxiliary extension problem. Let C be a finite set of allowed parts and let h be fixed. What conditions preserve a shift inequality if arbitrarily many large parts are added? For integer $s \ge -h$, put $g(s) = P_C(h)P_C(s) - P_C(s + h)$. Suppose the negative mass $B = \sum_{s \ge -h} \max\{-g(s), 0\}$ is finite. If an integer $J > h$ satisfies $g(s) > B$ for all $s \ge J - h$, then for every $D \subseteq \{J, J + 1, \dots\}$ disjoint from C and all $n \ge J - h$,

$$P_{C \cup D}(h)P_{C \cup D}(n) > P_{C \cup D}(n + h). \tag{R}$$

Indeed $P_{C \cup D}(h) = P_C(h)$ and the difference in (R) is the sum of $g(n - |\lambda|)$ over all D-partitions λ of weights up to $n + h$. A negative term must have nonempty λ, since the empty partition gives $g(n) > B$. Delete its smallest part d. The resulting λ' contributes $g(s + d)$ where $s = n - |\lambda| \ge -h$. As $s + d \ge J - h$, this contribution exceeds B. For fixed λ' and fixed s, the deleted part is forced to be $n - |\lambda'| - s$. Thus the sum of all deficits assigned to one positive term is at most B. This leaves the empty-partition term strictly positive after all assignments and proves (R).

Theorem 3.1. *Roy's Conjecture 3.2 holds. In fact, in the eight strips $4 \le h \le 11$, the complete exception table remains valid when all squares at least 49 are replaced by any set of positive integer parts at least 49.*

Proof. Apply (R) to $C = \{1, 4, 9, 16, 25, 36\}$ and $J = 49$. For $n + h < 49$ the counts are those of C itself. The following table gives the negative mass B, the minimum of g on $s \ge 49 - h$, and

the complete non-strict cases for $n \geq h$ in each strip.

h	B	$\min g$	$P(h)P(n) = P(n+h)$	$P(h)P(n) < P(n+h)$
4	4	59	$5, 6, 7$	$\varnothing$
5	8	45	$5, 6, 8, 11, 15$	7
6	16	35	$8, 10, 12, 14$	$6, 7, 11, 15$
7	26	29	$8, 9, 12, 13, 19$	$7, 10, 11, 14, 15$
8	15	83	$\varnothing$	$\varnothing$
9	16	123	$\varnothing$	$\varnothing$
10	23	103	$\varnothing$	$\varnothing$
11	31	97	$\varnothing$	$\varnothing$

Here is a finite certificate for the infinite statements about g. Let $L = 3600$ and compute $p[0], \ldots, p[6L + 20]$ by starting with $p[0] = 1$ and zeros elsewhere. For each $a \in C$, in increasing order of n from a to $6L + 20$, add $p[n - a]$ to $p[n]$. This gives $p[n] = P_C(n)$ by multiplication of generating series. For each residue $0 \leq r < L$, $Q_r(k) = g(r + kL)$ is a polynomial in k of degree at most five on all $k \geq 0$. Indeed,

$$\prod_{a \in C} (1 - z^a)^{-1} = (1 - z^L)^{-6} \prod_{a \in C} (1 + z^a + \cdots + z^{L-a}).$$

The numerator has degree less than $6L$. The contribution of a monomial in residue class r involves $\binom{k-j+5}{5}$ with $0 \leq j \leq 5$. When $0 \leq k < j$ this polynomial is zero, so validity extends all the way to $k = 0$. The shift by h in g preserves this polynomiality. Compute $Q_r(0), \ldots, Q_r(5)$ from the $p[n]$. Successively replacing adjacent entries by their forward differences gives $\Delta^j Q_r(0) \geq 0$ for $1 \leq j \leq 5$ for every h, r listed. The binomial-basis expansion of Q_r therefore makes it nondecreasing for all $k \geq 0$. Its values on $L \leq s < 2L$ all exceed the displayed B. It now suffices to sum the deficits for $-h \leq s < L$ and take the minimum for $49 - h \leq s < 2L$ to reproduce the two columns. All these exact integer operations are provided in the accompanying short verifier, `roy_square_verify.py`.

For $h = 1, 2, 3$ one has $P(h) = 1$. The increment $P(j) - P(j - 1)$ counts the square partitions of j lacking a part 1. Every $j \geq 24$ is a sum of fours and nines: the representatives $0, 9, 18, 27$ for the four residue classes suffice. Consequently $P(n + h) > P(n)$ for all $n \geq 23$. A direct recurrence up through 25 gives equality, for $h = 1$, at $n = 1, 2, 4, 5, 6, 9, 10, 13, 14, 18, 22$, for $h = 2$ at $n = 4, 5, 9, 13$, and for $h = 3$ at $n = 4$. Every other pair in these strips fails. Gajdzica's theorem covers $h \geq 12$. Comparing these disjoint ranges with the published table proves the conjecture. □

4 Ultra-log-concavity for dominating sets in powers of paths and cycles

Galvin and Zhang, *The domination polynomial of powers of paths and cycles*, arXiv:2408.12731, prove unimodality and ask in Conjecture 1.8 whether both coefficient sequences are ultra-log-concave for every order n and power $\ell \geq 1$. For a graph on n vertices set $f_k = d_k / \binom{n}{k}$, where d_k counts its k-element dominating sets. Their conclusion asks for $f_k^2 \geq f_{k-1} f_{k+1}$, not merely log-concavity of d_k.

Theorem 4.1. *For every $\ell \geq 1$, the ordinary domination sequences of P_n^ℓ and C_n^ℓ are ultra-log-concave wherever the corresponding graphs are defined.*

A comparison between bounded compositions. Let $U_b(x) = x + x^2 + \cdots + x^b$ for $b \geq 1$. Fix an integer N and a nonnegative finite sequence $v = (v_j)$. Consider a random integer with probabilities proportional to $v_j h_{N-j}$, where h is the coefficient sequence of a product of

polynomials U_b, allowing the empty product. Replacing h by the coefficients $h' = h * [x^\bullet]U_c$ for any $c \geq 1$ makes the random integer stochastically smaller, whenever both distributions can be normalized.

Indeed h is log-concave with interval support. To recall the convolution argument, the Toeplitz matrix $(h_{u-v})_{u,v\in\mathbb{Z}}$ has all order-two minors nonnegative exactly when its interval-supported sequence is log-concave (multiply consecutive ratio inequalities). Interval indicators have this property. Their convolutions do too by matrix multiplication and the order-two Cauchy–Binet identity; finite support makes the sums finite. The delta mass at zero also has the property. For $u_1 < u_2$ we consequently have

$$h'_{u_2}h_{u_1} - h'_{u_1}h_{u_2} = \sum_{i=1}^{c}\{h_{u_2-i}h_{u_1} - h_{u_1-i}h_{u_2}\} \geq 0.$$

This comparison includes the zeros beyond the supports. Put $u = N - j$. For $j \leq t < j'$ it gives $(v_j h'_{N-j})(v_{j'}h_{N-j'}) \geq (v_j h_{N-j})(v_{j'}h'_{N-j'})$. Summing over these pairs and normalizing shows that the probability of $j \leq t$ increases.

Deletion probabilities. In any nonempty graph the indices with $f_k > 0$ form a terminal interval. Given a uniform dominating k-set choose one of its members uniformly and delete it. The probability that the remaining set dominates is

$$s_k = \frac{(n-k+1)d_{k-1}}{kd_k} = \frac{f_{k-1}}{f_k}.$$

Every dominating $(k-1)$-set indeed has all $n-k+1$ possible one-element extensions. It suffices to show s_k increases weakly throughout the positive range. For $k = 1$ the deletion probability is zero. Empty graphs have a single coefficient.

Put $Q = 2\ell + 1$. On the cycle choose a distinguished selected vertex uniformly and list the successive clockwise distances between selected vertices, starting there. The ordered k distances are uniform among compositions of n with all parts in $[1, Q]$: each feasible ordered tuple and starting vertex gives exactly one pair of a dominating set and its distinguished member. Bounding each part by Q is equivalent to covering every vertex between consecutive selected vertices. For $k \geq 2$, deleting the distinguished member preserves domination exactly when the sum j of the two incident gaps is at most Q. Its law has weights

$$([x^j]U_Q^2)([x^{n-j}]U_Q^{k-2}).$$

Going from k to $k+1$ multiplies the residual polynomial by U_Q. The comparison just proved makes j stochastically smaller. Hence $s_{k+1} \geq s_k$ on the cycle, including the case $k = 2$.

On the path with vertices $1, \ldots, n$, for selected positions $x_1 < \cdots < x_k$ form the end gaps $x_1, n+1-x_k$ and interior gaps $x_{i+1} - x_i$. Their sum is $n+1$. Domination is equivalent to both end gaps being at most $A = \ell + 1$ and all interior gaps being at most Q. These positive gap vectors are uniform at fixed k since each gives one subset. For $k \geq 2$ let u_k be the probability that the first selected vertex can be deleted. This is the probability that the first end gap plus the first interior gap is at most A. The last selected vertex has the same probability by reflection. For $k \geq 3$ let v_k be the corresponding probability for an interior selected vertex. Exchangeability of the interior gaps makes this independent of the interior vertex index. Its condition is that two adjacent interior gaps sum to at most Q.

For each type of pair, its weight at a sum j is the coefficient at j of the product of its two gap polynomials, multiplied by the coefficient at $n+1-j$ of all the other gap polynomials. Passing from k to $k+1$ multiplies only the latter product by U_Q. Consequently $u_{k+1} \geq u_k$ for $k \geq 2$, and $v_{k+1} \geq v_k$ for $k \geq 3$ whenever the probabilities are defined.

There is also a comparison at the same $k \geq 3$. Write a for the left end gap, and b, c for the first two interior gaps. Conditional on b, all the other gaps apart from a, c, and the resulting sum $a + c = S$, the two-part split is uniform over $1 \leq a \leq A$, $1 \leq c \leq Q$. The random variables $A - a$ and $Q - c$ are uniform on integer intervals of equal length, since $a \mapsto c = S - a$ is a bijection. Their least possible values are $\max(0, A + 1 - S)$ and $\max(0, Q + 1 - S)$, respectively. Since $Q \geq A$, $Q - c$ stochastically dominates $A - a$. Comparing the probabilities that their values are at least b proves

$$\mathbb{P}(b + c \leq Q) \geq \mathbb{P}(a + b \leq A), \qquad v_k \geq u_k.$$

At $k = 2$ one has $s_2 = u_2$. For $k \geq 3$ one has $s_k = 2u_k/k + (1 - 2/k)v_k$. Both probabilities increase with k and the weight of the larger one increases, so $s_{k+1} \geq s_k$. At the transition to $k = 3$ the same conclusion follows from $s_3 \geq u_3 \geq u_2$. This proves Theorem 4.1.

5 Deleting one leaf can move a tree's domination mode arbitrarily far

S. Zhang, *Domination polynomial is unimodal for large graphs with a universal vertex*, arXiv:2111.00641, Conjecture 4.1, predicts that deleting a leaf changes the mode of a tree's domination polynomial by at most one whenever both polynomials are unimodal. This is the all-tree question of Burcroff and O'Brien, arXiv:2110.00709, Question 8.1. Their separate Conjecture 3.7 in that preprint concerns short-legged spiders.

Theorem 5.1. *There are trees T and leaves v for which both ordinary domination polynomials are unimodal and every mode of $D(T - v, x)$ exceeds every mode of $D(T, x)$ by an arbitrarily large amount.*

The tree and its exact polynomial. For $d \geq 1$ put $K = 2 \cdot 3^d$. At a vertex r attach b private leaves, where $b = 1, 2$, and K neighbors u_i. Attach to each u_i d disjoint paths $u_i - t_{ij} - w_{ij}$ of length two. Write T_b for this tree. Deleting a private leaf of r from T_2 gives T_1. With

$$A(x) = 2x + x^2, \qquad P(x) = (1 + x)A(x)^d, \qquad Q(x) = P(x) - x^d,$$

the polynomial is exactly

$$D_b(x) = x(1 + x)^b P(x)^K + x^b Q(x)^K.$$

Indeed at least one vertex of each pair $\{t_{ij}, w_{ij}\}$ must be selected. If r is selected the choice of u_i is free. Otherwise all b private leaves are selected and dominate r; the excluded choice in each branch is precisely to omit u_i and all t_{ij} and select all w_{ij}.

For any nonzero polynomial L with nonnegative coefficients, let X_L denote the integer random variable with probability generating function $L(x)/L(1)$. The law of X_P is $d + \mathrm{Bin}(d, 1/3) +$ Bernoulli$(1/2)$ with independent terms. The law of X_Q results from removing mass $1/K$ at d and renormalizing. Thus for $m_P = \mathbb{E}X_P$ and $m_Q = \mathbb{E}X_Q$,

$$m_P = \tfrac{4}{3}d + \tfrac{1}{2}, \qquad m_Q - m_P = \frac{d/3 + 1/2}{K - 1}.$$

The two summands of $D_b(1)$ have masses proportional to 2^b and $h_K = (1 - 1/K)^K$ respectively. Writing $\alpha_b = h_K/(2^b + h_K)$ gives the mean size μ_b under the uniform law on dominating sets:

$$\mu_b = (1 - \alpha_b)(Km_P + 1 + b/2) + \alpha_b(Km_Q + b).$$

In particular

$$\mu_1 - \mu_2 = \frac{d}{3}\left(\frac{e^{-1}}{2 + e^{-1}} - \frac{e^{-1}}{4 + e^{-1}}\right) + O(1). \qquad (*)$$

Below we show that D_b is unimodal and every mode is within $O(1)$ of μ_b, uniformly as $d \to \infty$.

The two summand distributions. Let $p(k)$ stand for the normalized coefficients of either summand. Its law is a sum of K independent branch sizes (X_P or X_Q) and one independent bounded variable encoding the root and private leaves. Write ν, V, κ_3 for its mean, variance and third cumulant. For each fixed $C > 0$, uniformly when $|k - \nu| \le Cd$,

$$\sqrt{2\pi}V^{3/2}\,[p(k+1) - p(k)] = \nu - k - \tfrac{1}{2} - \frac{\kappa_3}{2V} + o(1), \tag{5.1}$$

$$\sqrt{2\pi}V^{3/2}\,[p(k+1) - 2p(k) + p(k-1)] = -1 + o(1). \tag{5.2}$$

In both distributions

$$V = V_0 + O(d^2), \qquad V_0 = K(2d/9 + 1/4), \qquad |\kappa_3| = O(Kd).$$

To see this, removing probability $1/K$ from a law supported on $[d, 2d+1]$ changes its first three raw moments by $O(d^j/K)$ at order j. Hence its second and third cumulants change by $O(d^2/K)$ and $O(d^3/K)$, respectively. The Bernoulli sum for X_P has variance $2d/9 + 1/4$ and third cumulant $O(d)$; cumulants add.

We give details of the local estimates because a weak normal limit alone does not locate a mode to bounded precision. The single-branch characteristic function for X_P has modulus at most e^{-cdt^2} on $|t| \le \pi$, by $|1-p+pe^{it}|^2 = 1-2p(1-p)(1-\cos t)$. For X_Q the modulus is at most $(Ke^{-cdt^2}+1)/(K-1)$. Its Kth power is bounded by $C_1 e^{-c_1 K \min(dt^2,1)}$: factor out $(K+1)/(K-1)$ and use $1 - e^{-cdt^2} \ge c_2 \min(dt^2, 1)$. The bounded independent factor has characteristic modulus at most one. Thus the Fourier integral over $K^{-2/5} < |t| \le \pi$ is negligible even after multiplication by any fixed power of K.

On the remaining interval, the characteristic function of the centered sum has expansion

$$e^{-Vt^2/2}\left(1 + \frac{\kappa_3(it)^3}{6} + O\big(Kd^4|t|^4 + (Kd)^2|t|^6\big)\right).$$

For justification, a centered branch is bounded by $2d+1$. On a complex disk of radius c_3/d about zero its characteristic function is within $1/2$ of one for sufficiently small c_3. The analytic logarithm is bounded there. Its Taylor remainder after order three is $O(d^4|t|^4)$ by Cauchy's estimate on half the disk. Sum these logarithms, including the bounded independent factor, and exponentiate. On $|t| \le K^{-2/5}$ the cubic term and the fourth-order remainder both tend uniformly to zero.

Inverting for the forward difference introduces the multiplier

$$(e^{-it} - 1)e^{-it(k-\nu)} = -it - (k - \nu + 1/2)t^2 + O((1 + |k-\nu|^2)|t|^3).$$

The odd leading term integrates to zero against the centered Gaussian. Its product with the cubic term contributes $-\kappa_3 t^4/6$. Integrating the remaining even terms and using $\int_{\mathbb{R}} t^4 e^{-Vt^2/2}\,dt / \int_{\mathbb{R}} t^2 e^{-Vt^2/2}\,dt = 3/V$ yields (5.1). For example, after normalization the multiplier remainder is $O((1+d^2)/\sqrt{V})$ and the characteristic remainder times $O(|t|)$ contributes $O(Kd^4/V^{3/2} + (Kd)^2/V^{5/2})$, both $o(1)$. The product of the quadratic multiplier term and the cubic characteristic term is likewise $O((1+d)Kd/V^{3/2}) = o(1)$ after normalization. The omitted Gaussian tails are exponentially small. The central second difference uses $-t^2 + O((1+|k-\nu|)|t|^3)$ as multiplier; the same bounds give (5.2).

Unimodality and the mode displacement. Each summand has log-concave coefficients and interval support. Indeed P is, up to a monomial, a product of positive linear polynomials, and Q differs only in a reduced lowest nonzero coefficient; this reduction preserves log-concavity and positivity of that coefficient. Convolution preserves log-concavity of nonnegative interval-supported sequences, as shown in the proof of Theorem 4.1 (or by the order-two Toeplitz minors).

Consequently (5.1) brackets all modes of each summand between its mean minus and plus an absolute constant, by checking the sign of the first difference on either side.

The two means differ by $O(d)$. Below a common central interval containing both mode ranges the mixture increases, and above it the mixture decreases, since its two summands do so individually. Throughout that interval the mixture is strictly concave in the discrete sense by (5.2), proving unimodality. To locate its own modes, multiply its forward difference by $\sqrt{2\pi}V_0^{3/2}$. Applying (5.1) to the two summands and averaging gives $\mu_b - k + O(1)$ uniformly there. Indeed $V_0^{3/2}/V^{3/2} = 1 + O(d^2/V_0)$, and its error even when multiplied by $O(d)$ tends to zero. Thus every mode is $\mu_b + O(1)$. Equation (*) now proves the theorem.

6 The generalized gamma convolution range of the Kanter variable

For $0 < \beta < 1$, the Kanter variable K_β is the positive factor in the independent factorization $Z_\beta \overset{d}{=} L^{1-1/\beta}K_\beta$, where L is unit exponential and $\mathbb{E}e^{-sZ_\beta} = e^{-s^\beta}$. Equivalently its Mellin transform is given in (5) of Hasebe, Simon and Wang, *Some properties of the free stable distributions*, Ann. Inst. H. Poincaré Probab. Statist. 56 (2020), 296–325, arXiv:1805.01133. A generalized gamma convolution (GGC) is a weak limit of finite sums of independent gamma variables with arbitrary scales.

Theorem 6.1. *The law of K_β is a GGC if and only if $\beta \le 4/5$. Thus Conjecture 1 of Hasebe, Simon and Wang holds.*

They prove necessity in Corollary 2 and sufficiency for $\beta \le 3/4$ via Proposition 1. Their Thorin phase criterion (13)–(14) reduces the outstanding interval to the following problem. Put $a = 1 - \beta$, $1/5 \le a \le 1/4$ and $x = r^{1-a} > 0$. Define

$$R(x) + iI(x) = \sum_{n\ge 0} \frac{e^{i\pi na}x^n}{n!\,\Gamma(1-na)}.$$

The Thorin phase is $\pi^{-1}\arg(R+iI)$. Its nondecrease on the whole positive half-line is equivalent, by the source, to the GGC property. We prove $R > 0$ and $(I/R)' > 0$ throughout this strip. The reflection formula gives, for positive n,

$$[x^n]R = \frac{\Gamma(na)\sin(2\pi na)}{2\pi n!}, \qquad [x^n]I = \frac{\Gamma(na)\sin^2(\pi na)}{\pi n!}.$$

In particular $I > 0$. The positive coefficients will control I in the compact part of the argument. The oscillating coefficients of R require a different estimate for interpolation.

Integral and contour. Expanding exponentials under the gamma integral gives

$$R = 1 + \frac{1}{2\pi}\Im\int_0^\infty e^{-t}(e^{xe^{2\pi ia}t^a} - 1)\,\frac{dt}{t},$$
$$I = \frac{1}{2\pi}\int_0^\infty e^{-t}(e^{xt^a} - \Re e^{xe^{2\pi ia}t^a})\,\frac{dt}{t}.$$

In the first integral one can take the imaginary part of the unsubtracted integrand with a lower cutoff tending to zero. Use the principal power and deform to the path from 0 to iS and then horizontally to $+\infty + iS$. The small circular arc for the unsubtracted integrand contributes $+1/4$ to R. The vertical connector at the right endpoint tends to zero since $t^a = o(|t|)$. On the vertical segment $t = iu$ the exponential $e^{-t+xe^{2\pi ia}t^a}$ has modulus at most one, because $\phi = 5\pi a/2 \in [\pi/2, 5\pi/8]$. At $t = u + iS$ its real exponent has derivative at most

$-1 + axS^{a-1}$. Thus when $axS^{a-1} < 1/2$ the horizontal exponential has modulus at most $e^{-u/2}$. Its contribution from the unsubtracted integrand, and that to the first x derivative, tend to zero as $S \to \infty$ (bounded by constants times S^{-1} and S^{a-1}). This argument justifies the full vertical representation even at $a = 1/5$ where it is oscillatory.

Unbounded range. Put $p = 1/a$, $T = (ax)^{1/(1-a)}$, $v_0 = T^a$. If $x \geq 37$ then $T = (x/p)^{p/(p-1)} \geq (37/5)^{5/4} > 12$, since the expression decreases in p here. Let $q = \sin\phi > 0.92$, $A = Tq^{p/(p-1)} > 10$, $v_* = v_0 q^{1/(p-1)}$, and $D = -pA\cot\phi \geq 0$. On the vertical ray substitute $t = iAz^p$. The formula for R becomes

$$R = \frac{5}{4} + \frac{p}{2\pi}\Im\int_0^\infty e^{-Dz+iA(pz-z^p)}\,\frac{dz}{z}.$$

Replacing $pz - z^p$ by pz gives imaginary integral $\arg(D + ipA) = \pi - \phi$, by the elementary Laplace sine integral (including $D = 0$ by a limit). The corresponding value of R is $p/2$. We give an explicit upper bound 0.8 for the error in the imaginary integral. Throughout, a one-sided oscillatory integral with monotone phase derivative of magnitude at least b has modulus at most $2/(Ab)$ when the exponential is $e^{iA\,\text{phase}}$. Integration by parts proves this inequality. Multiplication by a nonnegative decreasing amplitude costs only its maximum, by averaging indicators of initial subintervals.

Take $c = 3/5$ and $y = z - z^p/p$ on $0 < z < c$. Compare the integrand there with the linear-phase integral up to $y(c)$. The absolute discrepancy of the measures is at most

$$-\log(1 - c^{p-1}/p) + \frac{c^{p-1}}{ep(p-1)(1 - c^{p-1}/p)} < 0.065.$$

Indeed $0 < y'/y \leq 1/z$ and the integral of $1/z - y'/y$ is the first term. The damping error is bounded using $|e^{-Dz} - e^{-Dy}| \leq (z-y)/(ey)$ and $y'/y \leq 1/z$. The remaining linear tail from $y(c)$ has modulus at most $2/[pAy(c)] < 0.09$. For the nonlinear tail split at $7/8$ and $9/8$. The absolute bounds for the three pieces are respectively

$$\frac{2}{cpA(1-(7/8)^{p-1})}, \qquad \log(9/7), \qquad \frac{2}{(9/8)pA((9/8)^{p-1}-1)}.$$

Their sum is below 0.62, giving the claimed error. For R' use the differentiated contour before rescaling, which replaces dz/z by $e^{i\phi}v_* dz$. The derivative bound instead uses the three intervals from zero to $7/8$, from $7/8$ to $9/8$, and from $9/8$ onwards. It yields

$$\frac{|R'|}{v_0} \leq \frac{p}{2\pi}\left(\frac{2}{pA(1-(7/8)^{p-1})} + \frac{1}{4} + \frac{2}{pA((9/8)^{p-1}-1)}\right) < 0.09p.$$

Thus $R > p(1/2 - 0.8/(2\pi)) > 0$ and $R'/R < v_0/4$.

To bound I'/I, use $t = Tz^p$ on the positive real ray, for which $xt^a = pTz$. Let $u = pTz$ and $\eta = 2\pi a$. Apart from the common factor $pe^{-Tz^p}/(2\pi)$ and the integration in z, the integrands of I and I'/v_0 are

$$\frac{e^u - e^{u\cos\eta}\cos(u\sin\eta)}{z}, \qquad e^u - e^{u\cos\eta}\cos(u\sin\eta + \eta).$$

Both are nonnegative. If $z \geq 3/8$ then $e^{-u(1-\cos\eta)} \leq e^{-12}$ as $\cos\eta < 1/3$. The second integrand is at least $(3/8)(1-e^{-12})/(1+e^{-12})$ times the first. The portion of I below $3/8$ is less than $1/100$ of its total. Indeed, omitting the common constant $p/(2\pi)$, that portion is at most $(3/4)pTe^{(3/8)pT}$ since $|e^u - e^{ue^{i\eta}}| \leq 2ue^u$. The portion on $[9/10, 1]$ is at least $0.09e^{(0.9p-1)T}$. Their ratio is at most $(0.75p/0.09)Te^{-(0.525p-1)T} < 0.01$ for $T \geq 12$ and $4 \leq p \leq 5$. Therefore $I'/I > 0.3v_0 > R'/R$. This proves strict phase increase on the unbounded range.

A compact uniform certificate. For $0 \le x \le 37$ and $1/5 \le a \le 1/4$, the rotated contour at $S = 128$ gives

$$|R_{xx}| \le 8, \quad |R_{xxx}| \le 16, \quad |R_{aa}| \le 2 \cdot 10^6, \quad |R_{xaa}| \le 6 \cdot 10^6.$$

Here are details for these deliberately loose constants. Set $H = e^{2\pi i a} t^a$, $L = 2\pi i + \log t$. In the unsubtracted integral the four derivative multipliers are $H^2, H^3, (xH + x^2H^2)L^2$, and $(H + 3xH^2 + x^2H^3)L^2$. On the vertical segment $|L|^2 = (\log u)^2 + 25\pi^2/4$. Thus the integral of $|H|^m|L|^2$ with measure $|e^{-t+xH}dt/t|/(2\pi)$ is at most

$$\frac{1}{2\pi}\left[\frac{2}{(m/5)^3} + \frac{62}{m/5} + \frac{89\,128^{m/4}}{m/5}\right]$$

on this segment, by splitting at $u = 1$. Without $|L|^2$ its bound is $128^{m/4}/(2\pi m/5)$. On the horizontal part $|t|^{ma-1} \le 1$ for $1 \le m \le 3$, $|L| \le 13 + u/128$ and the damping is $e^{-u/2}$, so the further contributions are at most 60 with $|L|^2$ and $1/\pi$ without it. These bounds imply the four constants.

The grid has step $1/8$ in x and $1/2000$ in a. At its vertices evaluate R, R' through $n = 383$ by 176-bit Arb arithmetic. The error of bilinear interpolation of either function is at most $(M_{xx}/8^2 + M_{aa}/2000^2)/8$, where M_{xx} and M_{aa} bound its respective pure second derivatives. This follows by applying the one-dimensional linear interpolation error successively in the two variables. On each order interval enclose the positive coefficients $i_n = [x^n]I$ above and below. On a grid cell a lower bound for xI'/I is given by evaluating $\sum n i_n x^{n-1}$ from below at the left endpoint and $\sum i_n x^{n-1}$ from above at the right. This expression also has a positive limiting value at $x = 0$.

A tail allowance 10^{-50} suffices for each truncated series involved. In fact the four terms $n = 4l, \dots, 4l+3$ with $l \ge 96$ contribute at most $4(4l+4)(l+1)!37^{4l+3}/(4l)!$, including a factor n when necessary. The ratio of successive majorants is below $1/2$, using $(l+2)/(4l+1)^4$ decreasing there. The first majorant is below $5 \cdot 10^{-51}$. The verifier `kanter_verify.py` checks $R > 0$ and $(xI'/I)R > xR'$ for all 29,600 closed grid cells by these bounds. For $x > 0$ this implies $(I/R)' > 0$, completing the proof.

Consequences for stable laws. In the notation of the source, the one-sided free stable law X_β is a classical GGC for every $0 < \beta \le 4/5$. Indeed, (3) expresses it as the independent product of K_β and $U^{1-1/\beta}$ with U uniform on $(0,1)$. The GGC implication for that product is precisely the reduction in the source's Section 2.3.1. Its Section 2.3.2 then gives the extended Thorin property for the two-sided $X_{\beta,\rho}$, $0 \le \rho \le 1$, throughout the same order range. These are direct consequences of Theorem 6.1. For the ordinary positive stable law, equation (12) and Remark 4(a) of the source also identify $[1/5, 1)$ as the exact range on which the survival function of $Z_a^{-a/(1-a)}$ is hyperbolically completely monotone.

7 A shifted free stable law that is not a gamma mixture

For $0 < \beta < 1$ let X_β be the positive free stable variable in the normalization of Hasebe, Simon and Wang, and let $b_\beta = \inf \operatorname{supp} X_\beta$. In Remarks 2(b) and 10(b) of *Some properties of the free stable distributions* the authors predict that $Y_\beta = X_\beta - b_\beta$ is a scale mixture of gamma distributions of shape $3/2$ for all $0 < \beta < 1$. Both remarks concern the same conjecture.

Theorem 7.1. *Y_β is a gamma scale mixture with shape $3/2$ for $0 < \beta \le 4/5$ and is not such a mixture for $23/25 \le \beta < 1$. In particular the proposed all-order-parameter assertion is false.*

The intermediate interval $(4/5, 23/25)$ is not classified here. For the positive range, the phase limit in Proposition 5 of the source is $1/2$. With Theorem 6.1, the argument of its Remark 10(b)

applies: the Thorin mass of K_β is $1/2$, so $K_\beta - b_\beta$ is a gamma scale mixture of shape $1/2$. The same remark proves the implication from this last property to the required shape-$3/2$ mixture of $X_\beta - b_\beta$.

Proof of failure near order one. If the gamma mixture representation held, then $g_\beta(x) = x^{-1/2} f_{Y_\beta}(x)$ would be completely monotone. In fact mixing the gamma rates averages positive multiples of e^{-tx} after division by $x^{1/2}$. Proposition 6 of the source gives an all-orders expansion $g_\beta(x) \sim \sum_{n\ge 0} a_n(\beta) x^n$ at $0+$. Its formula implies that $(-1)^n a_n(\beta)$ has the sign of

$$A_n(\beta) = [v^{2n}](1-v)^{-2}\,{}_2F_1(1+\beta, 1; 3; v)^{-n-1/2}.$$

If g_β is CM then $(-1)^n a_n(\beta) \ge 0$ for all n. Indeed g_β has a finite limit at zero. The nth forward difference at zero of each e^{-tx}, multiplied by $(-1)^n$, is nonnegative. By the positive Laplace representation the same is true for g_β. Dividing by the nth power of the mesh and taking the mesh to zero gives $(-1)^n n! a_n(\beta) \ge 0$ directly from the expansion. Termwise differentiation of an asymptotic series is unnecessary.

For the eight closed intervals $[i/100, (i+1)/100]$ with $i = 92$ to 99, respectively, the following orders certify $A_n(\beta) < 0$:

$$(71, 55, 45, 35, 30, 29, 24, 23).$$

Here is the finite rational calculation. Set $\beta = (i+u)/100$ and $N = 2n$, $d = -n - 1/2$. The hypergeometric coefficients are $b_0 = 1$ and $b_k = b_{k-1}(\beta + k)/(k+2)$. For the coefficients of its dth power put $h_0 = 1$ and use logarithmic differentiation:

$$h_k = \frac{1}{k}\sum_{l=1}^{k}((d+1)l - k) b_l h_{k-l}, \qquad A_n = \sum_{k=0}^{N}(N+1-k)h_k.$$

Write $A_n((i+u)/100) = \sum_{l=0}^{N} d_l u^l$. Its degree-$N$ Bernstein coefficients on $[0,1]$ are $\sum_{l=0}^{j} d_l \binom{j}{l}/\binom{N}{l}$, $0 \le j \le N$. The verifier `mixture_rational.py` checks that every one is strictly negative in each row. The Bernstein basis is nonnegative and sums to one. This proves strict negativity on the entire union of intervals and contradicts the necessary sign for the original mixture. □

The mixture obstruction tests the density at the left edge. The GGC criterion in Section 6 instead uses the phase of a Laplace transform of a different variable. The two conditions lead to separate conclusions in the source's terminology.

8 Shifted gamma saddles and the stable-edge coefficient identity

In Remark 12 of *Some properties of the free stable distributions* (Hasebe, Simon and Wang, arXiv:1805.01133), the authors conjecture an all-orders coefficient formula for the density of their endpoint variable Y_1. Proposition 7 identifies its coefficient $a_n(1)$ with

$$\frac{(-1)^n 2^{2n+1/2} n!}{\pi(2n+1)!}(c_0 + \cdots + c_n),$$

where c_n are the Stirling coefficients of the gamma function. Their conjecture is that the following b_n equals this partial sum:

$$b_n = \frac{1}{2^{2n+1/2} n!}\,\frac{d^{2n}}{dv^{2n}}\left\{\frac{1}{(1-v)^2}\left(\frac{v^2}{-v - \log(1-v)}\right)^{n+1/2}\right\}\Bigg|_{v=0}.$$

All analytic fractional powers here have their positive value at zero. We derive the identity from the shifted gamma integral, keeping a real shift d as a parameter.

Proposition 8.1. *For $n \geq 0$ put*

$$B_n(d) = (2n-1)!![v^{2n}](1-v)^{-d-1}\left(\frac{v^2}{2(-v-\log(1-v))}\right)^{n+1/2}, \qquad (-1)!! = 1.$$

These are polynomials of degree exactly $2n$ in d, with $B_n(0) = c_n$. They satisfy $B_0(d) = 1$ and $B_n(d+1) - B_n(d) = (d+1)B_{n-1}(d+1)$ for $n \geq 1$. In particular $b_n = B_n(1) = \sum_{j=0}^{n} c_j$ and hence the proposed stable-edge formula holds in every order.

Proof. For $x > \max(d, 0)$ apply the substitution $t = x(1-v)$ in the integral for $\Gamma(x-d)$. The normalized integral becomes

$$G_d(x) = \sqrt{\frac{x}{2\pi}}\,\frac{e^x\Gamma(x-d)}{x^{x-d}} = \sqrt{\frac{x}{2\pi}}\int_{-\infty}^{1}(1-v)^{-d-1}e^{-x\varphi(v)}\,dv, \qquad \varphi(v) = -v - \log(1-v).$$

The phase is strictly positive away from its unique minimum at zero, where $\varphi(v) = v^2/2 + O(v^3)$. Contributions outside a fixed neighborhood of zero are exponentially small: on that complement bound $e^{-x\varphi}$ by $e^{-(x-x_0)\inf\varphi}e^{-x_0\varphi}$ for a fixed $x_0 > \max(d, 0)$. In that neighborhood use the analytic coordinate $w = v\sqrt{2\varphi(v)/v^2}$, with inverse $v(w)$. Expand $A(w) = (1 - v(w))^{-d-1}v'(w)$ and integrate symmetrically. The Gaussian moments give $G_d(x) \sim \sum_{n\geq 0} x^{-n}(2n-1)!![w^{2n}]A(w)$; the Taylor remainder through odd order $2N+1$ is $O(x^{-N-1})$. By the change of variable in a coefficient residue,

$$[w^{2n}]A(w) = [v^{2n}](1-v)^{-d-1}\big(v/w(v)\big)^{2n+1}.$$

This yields the displayed coefficient, whose term of degree $2n$ in d comes only from $[v^{2n}](1-v)^{-d-1}$ and has nonzero multiplier. At $d = 0$ the normalized integral defines the ordinary Stirling coefficients. Finally $G_{d+1}(x) = G_d(x)/(1 - (d+1)/x)$ by the gamma recurrence. Comparison of coefficients proves the finite-difference relation. At $d = 1$ the factorial conversions in the proposition give $B_n(1) = b_n$. Putting $d = 0$ in the recurrence gives the claimed partial sums. □

This derivation uses classical saddle-point analysis and coefficient inversion. It avoids interchanging the limiting stable law with its asymptotic expansion. Generalized Stirling coefficients with a shift have long been studied, which limits the novelty attributable to the adjacent coefficient identity.

9 A sharp parameter half-plane for Jacobi endpoint comparison

Booker, Strömbergsson and Then, *Bounds and algorithms for the K-Bessel function of imaginary order*, LMS J. Comput. Math. 16 (2013), 78–108, Conjecture 1, equation (34), consider $f_n(x, s) = n!P_n^{(-s,s)}(x)$. They establish the endpoint bound for $j = 0$, for $r = 0$, and for $n \leq j + 100$. The following classification in asymmetric weights includes all the derivative orders in their question.

Theorem 9.1. *Fix $a, b > -1$. For nonnegative integers m, j, k put $Q(x, s) = P_m^{(a-s,b+s)}(x)$. The inequalities*

$$|\partial_s^j\partial_x^k Q(x, s)| \leq |\partial_s^j\partial_x^k Q(1, s)| \qquad (-1 \leq x \leq 1, \quad \Re s \leq 0)$$

hold simultaneously for all m, j, k if and only if $a \geq b$ and $a \geq -1/2$. In particular the 2013 conjecture holds at every $s = ir$, with no restriction on $n - j$.

Proof. First take $s = ir$, $j = k = 0$, and $m \geq 1$. Write $y(x) = Q(x, ir)$ and $L = m(m+a+b+1) > 0$. The Jacobi equation yields the exact energy identity

$$\frac{d}{dx}\{L|y|^2 + (1-x^2)|y'|^2\} = 2[(a+b+1)x + a - b]|y'|^2.$$

Suppose $a \geq b$ and $a \geq -1/2$. When $a+b+1 \geq 0$ the bracket is nondecreasing, so the maximum of the energy is at an endpoint. When $a+b+1 < 0$, its minimum is at $x = 1$, with value $2a+1 \geq 0$, and the same conclusion holds. The modulus of the endpoint at $+1$ is at least that at -1, since their values up to sign are

$$\frac{1}{m!}\prod_{h=1}^{m}(h+a-ir), \qquad \frac{1}{m!}\prod_{h=1}^{m}(h+b+ir)$$

and $h+a \geq h+b > 0$.

For a fixed x now view the endpoint and interior polynomials as functions of s, and write $P(s) = Q(1,s)$ and $H(s) = Q(x,s)$. The zeros of P lie strictly in the right half-plane. The Jacobi binomial formula gives the same degree-m leading coefficient $(-1)^m/m!$ for P and H (the two binomial factors contribute a binomial expansion of a constant). The preceding boundary estimate and the maximum modulus principle therefore give $|H/P| \leq 1$ throughout the closed left half-plane, using $H/P \to 1$ at infinity. For every $|u| < 1$, the polynomial $P + uH$ has degree m and all zeros in the open right half-plane. By Gauss–Lucas the same holds for each nonconstant derivative, and its mth derivative is nonzero. For any s in the closed left half-plane, $|H^{(j)}(s)| > |P^{(j)}(s)|$ would contradict this on taking $u = -P^{(j)}(s)/H^{(j)}(s)$. Higher derivatives vanish. This elementary use of polynomial comparison explains the transfer from the known zeroth derivative to all orders. For the x-derivatives use the Jacobi differentiation formula

$$\partial_x^k Q(x,s) = 2^{-k}(m+a+b+1)_k P_{m-k}^{(a+k-s,b+k+s)}(x), \qquad 0 \leq k \leq m.$$

Its prefactor is independent of s and the shifted parameters satisfy the same sufficient conditions. The constant case is immediate.

For necessity, $a < b$ fails already for $m = 1$, $x = -1$, $s = 0$. If $b \leq a < -1/2$, the bracket in the energy derivative is strictly negative on some interval $(c, 1]$. At $s = 0$, for sufficiently large m the derivative of $P_m^{(a,b)}$ has a zero x_m in $(c,1)$. Indeed, it is a nonzero multiple of the degree-$m-1$ orthogonal Jacobi polynomial for weight $(1-x)^{a+1}(1+x)^{b+1}$. Its largest zero tends to 1: the largest eigenvalue of the compression of multiplication by x to polynomials of degree at most $m-2$ is that zero, by the three-term recurrence. Its Rayleigh quotient on $(1+x)^{m-2}$ tends to 1 by concentration of the weighted integrals at the right endpoint. Integrating the strict energy decrease from x_m to 1 gives $|Q(x_m,0)| > |Q(1,0)|$, since the derivative term vanishes at both ends of this integration. This proves necessity. □

For $a = b = 0$, the hypothesis on Conjecture 1 in the source's Proposition 4 is consequently available for every positive integer n. The energy method and Gauss–Lucas comparison are classical. The full classification gives the precise obstruction to the same endpoint choice for complex-shift Jacobi weights.

10 The three global Bessel bounds in a 2009 conjecture

Baricz and Wu, *Sharp exponential Redheffer-type inequalities for Bessel functions*, Publ. Math. Debrecen 74 (2009), 257–278, concluding remark 6 of Section 1, conjectured full ranges for three inequalities. We retain their common source count. Let $j = j_{\nu,1}$ be the first positive zero of J_ν, let $y = x^2/j^2$, $R = (1-y)/(1+y)$, $F = \mathcal{J}_\nu(x) = 2^\nu\Gamma(\nu+1)x^{-\nu}J_\nu(x)$ (by its even series for real x), and write $F_+ = \mathcal{J}_{\nu+1}(x)$. Their proposals are $F \geq R$ for $-1 < \nu < \nu_0$, and $F \leq R^\beta$, $F_+/F \geq R^{-\gamma}$ for $\nu > -1$, throughout $0 \leq y < 1$. Here $\beta = j^2/[8(\nu+1)]$, $\gamma = \beta/(\nu+2)$, and $\nu_0 \in (1,2)$ is the unique order with $\beta = 1$.

Theorem 10.1. *The first inequality holds throughout the interval exactly when $-1 < \nu \leq 1/2$. The second holds throughout exactly when $\beta \leq 1$. The third holds throughout for $-1 < \nu \leq 1/2$,*

but fails for every sufficiently large ν, even at the fixed relative position $y = 255/256$. Thus all three proposed full ranges are false. The large-order failures of (1.2) and (1.3) also concern the ranges printed in Theorem 1 of the original article.

Proof. Set $r_k = j^2/j_{\nu,k}^2$, so that $r_1 = 1$, and $S = \sum r_k = j^2/[4(\nu+1)] = 2\beta$ by the first Rayleigh sum. The product formula gives $F = \prod_{k\ge1}(1 - r_k y)$. For the first inequality the log difference after removing $1 - y$ is

$$D(y) = \log(1+y) + \sum_{k\ge2} \log(1 - r_k y).$$

It is strictly concave through $[0, 1]$, and $D(0) = 0$. Consequently nonnegativity throughout is equivalent to $2\prod_{k\ge2}(1 - r_k) \ge 1$. Each r_k, $k > 1$, increases strictly with ν. In fact the Watson order derivative at any positive zero $c = j_{\nu,k}$ is

$$\frac{d}{d\nu} \log c = 2 \int_0^\infty K_0(2c \sinh v) e^{-2\nu v}\, dv$$

(DLMF 10.21.17), a strictly decreasing function of c for fixed ν. At $\nu = 1/2$ one has $r_k = 1/k^2$ and $\prod_{k\ge2}(1 - 1/k^2) = 1/2$ by telescoping. The convergent positive product is strictly decreasing in ν, by factorwise comparison, with a strict inequality for $k = 2$. This gives the first range.

For the second, $F/(1 - y)$ tends to a finite positive number as $y \uparrow 1$, ruling out $\beta > 1$. If $S \le 2$, then

$$-\log F - S \operatorname{artanh} y \ge -\log(1-y) + (S-1)y - S \operatorname{artanh} y \ge y - \log(1+y) \ge 0.$$

The middle expression decreases in S and was evaluated at $S = 2$.

For the third, logarithmic differentiation of the product in x and the Bessel recurrence give

$$F_+/F = \sum_{k\ge1} p_k (1 - r_k y)^{-1}, \qquad p_k = r_k/S.$$

Here $M = \sum p_k r_k = j^2/[4(\nu+1)(\nu+2)] = 2\gamma$, by the second Rayleigh sum or the linear coefficient of the quotient series in y. Whenever $\sum r_k^2 \le 2$ we have $M \le 2p_1$. Jensen's inequality and the same scalar estimate as above give

$$\log(F_+/F) - M \operatorname{artanh} y \ge -p_1 \log(1-y) + (M - p_1)y - M \operatorname{artanh} y \ge 0.$$

For $\nu \le 1/2$ one has $r_k \le 1/k^2$ and $\sum r_k^2 < 2$.

To see the opposite behavior, take ν large, $h = \nu + 1/2$. On $0 < x < h$ before the first zero let $u = J_{\nu+1}(x)/J_\nu(x)$. It satisfies $u' = 1 - 2hu/x + u^2$. The smaller root of this right-hand side is $b(x) = x/(h + \sqrt{h^2 - x^2})$, a strictly increasing function. Near zero, $u \sim x/(2\nu + 2) < b$. A first contact from below would give $u' = 0 < b'$, an impossibility. Thus $F_+/F \le 2(\nu+1)/(h + \sqrt{h^2 - x^2})$. The standard first-zero asymptotic $j_{\nu,1}/\nu \to 1$ (DLMF, Section 10.21(vii)) ensures $x = j\sqrt{255/256} < h$ eventually. The upper limit of F_+/F is at most $2/(1 + 1/16) < 2$ there, whereas $R^{-\gamma} \to 511^{1/8} > 2$. This proves the refutation. □

11 Fractional degrees of two trigamma remainders

F. Qi, *Completely monotonic degree of a function involving trigamma and tetragamma functions*, AIMS Mathematics 5 (2020), 3391–3407, Remark 7.4, conjectures degree four for each of two signed remainders. The same assertion appeared in the earlier preprint, equation (4.8) in version 3. Put

$$\Psi(x) = \psi'(x)^2 + \psi''(x), \qquad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}, \quad x > 0.$$

For the positive functions vanishing at infinity in this section, the degree is the supremum of the real r for which $x^r f(x)$ is completely monotone (CM), meaning $(-1)^j (x^r f(x))^{(j)} \ge 0$ for every $j \ge 0$. The source establishes CM of h_λ for $\lambda \le 0$ and of $-h_\mu$ for $\mu \ge 4$.

Theorem 11.1. *For every $\mu \geq 4$ the degree of $-h_\mu$ equals 3. The degree of h_0 equals 2. There is a continuous strictly increasing bijection $C : (0,1) \to (0,\infty)$, given below by an integral maximum, such that for $\lambda < 0$ the degree of h_λ equals $2 + C^{-1}(-\lambda/12)$. All these degrees are attained.*

Already the recurrence for the polygammas at zero gives

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(2\zeta(2) - \frac{37}{12} + \frac{\lambda}{6}\right) x^{-2} + O(x^{-1}).$$

A nonzero nonnegative decreasing function cannot tend to zero at the left endpoint. These expansions alone contradict the two proposed degrees of four. The exact replacement requires controlling the signs of fractional derivatives over a whole half-line.

Let $b(t) = t/(1-e^{-t})$, $K = b * b - tb$ with additive convolution, and define

$$V(t) = K''(t) - (t-2) - (2 + 13t/12)e^{-t},$$
$$W(t) = 1 - (11 + 9t)e^{-t}/12 - K'''(t), \qquad j(t) = 1 - (1+t)e^{-t}.$$

The trigamma transform $\psi' = \mathcal{L}b$ gives $\Psi = \mathcal{L}K$, and $K = O(t^3)$ at zero. Partial fractions and Laplace differentiation yield

$$x^2 h_\lambda = \mathcal{L}(V - \lambda j/12), \qquad -x^3 h_\mu = \mathcal{L}(W + (\mu - 4)te^{-t}/12).$$

Lemma 11.2. *$V(t), W(t) > 0$ for every $t > 0$. At zero $V(t) = t^2/6 + O(t^3)$. At infinity $V(t) \sim t^3 e^{-t}/6$ and $V'(t) \sim -t^3 e^{-t}/6$.*

Proof. The convolution identity

$$(1-e^{-s})^{-1}(1-e^{-(t-s)})^{-1} = \frac{(1-e^{-s})^{-1} + (1-e^{-(t-s)})^{-1} - 1}{1-e^{-t}}$$

gives the convergent expansion $K(t) = \sum_{k\geq 0} P_k(t) e^{-kt}$, where

$$P_k(t) = t^3/6 - t^2 + 2(\zeta(2) + H_k^{(2)})t - 4(\zeta(3) - H_k^{(3)}), \qquad H_k^{(r)} = \sum_{l=1}^{k} l^{-r}.$$

Indeed, in the convolution the remaining integral in the numerator is $2\int_0^t s(t-s)/(e^s - 1)\, ds$. Expand the denominator into exponentials and integrate to obtain the formula. The series and every fixed derivative converge absolutely locally on $t > 0$. Its first exponential implies the stated large-t asymptotics.

Here are reproducible sign bounds. For $t \geq 32$ one has $P_k \geq t^3/8$, $0 < P_k' \leq t^2/2 + 8$, $P_k'' = t - 2$ and $P_k''' = 1$. After the $k = 0$ cancellations, the kth polynomial contributing to V is $P_k'' - 2kP_k' + k^2 P_k$. This is at least $k^2(t^3/8 - t^2 - 16)$ for $k \geq 1$. For W the polynomial is $k^3 P_k - 3k^2 P_k' + 3kP_k'' - 1$, which is at least $k^3(t^3/8 - 3t^2/2 - 25)$. The $k = 1$ terms alone exceed the respective additional subtractions $(2 + 13t/12)e^{-t}$ and $(11 + 9t)e^{-t}/12$.

On $2 \leq t \leq 32$ apply the same expansion to $e^t V$ and $e^t W$, retaining $k \leq 32$. The absolute remainder of either is bounded by $2 \cdot 10^5 \, 33^3/7^{32}$. Indeed each polynomial in the preceding display is bounded absolutely by $10^5 k^3$ on this interval, and $e^{-t} < 1/7$. On $0 \leq t \leq 2$ compute the Taylor polynomials of V, W through degree 80 from

$$[t^{m+1}]K = \sum_{l=0}^{m} b_l b_{m-l} \frac{l!(m-l)!}{(m+1)!} - b_m, \qquad b_l = [t^l]b.$$

They have zero constant and linear terms, and $[t^2]V = 1/6$. On $|z| \leq 4$ one has $|b(z)| < 7$ by the even Bernoulli coefficient formula $|b_{2l}| = 2\zeta(2l)/(2\pi)^{2l}$ for $l \geq 1$. Integration along the straight segment gives $|K(z)| \leq 224$ there. On $|z| \leq 3$ the Cauchy derivative bounds for K'' and K''' and the explicit exponential terms bound $|V|, |W|$ by 2000. Hence the discarded tails after division by t^2 are at most $1500(2/3)^{81}$ on $[0,2]$. The verifier `trigamma_degrees.py` evaluates the two polynomial bounds and the finite exponential sums by outward-rounded interval arithmetic (100 bits). Bisection accepts respectively 11 and 166 boxes. □

An auxiliary transfer for fractional orders. For $0 \le s < 1$ define

$$(L_s u)(t) = \frac{1}{\Gamma(1-s)} \int_0^t (t-v)^{-s} u(v)\, dv.$$

For $0 < s < 1$ set

$$C(s) = \max_{t>0} \frac{-L_s V'(t)}{L_s j'(t)}.$$

The denominator is positive since $j'(t) = te^{-t}$. Notice that this is an explicit scalar optimization with a strictly invertible dependence on the degree. It involves no searches over derivative orders. Indeed, since $V(0) = j(0) = 0$, the Laplace transform of $L_s(V' + cj')$ is exactly $x^{2+s} h_{-12c}(x)$. By Bernstein's theorem and uniqueness of Laplace transforms this function is CM if and only if $c \ge C(s)$.

We verify all the asserted properties of the maximum. Write $R_s(t)$ for the quotient in its definition. As $t \downarrow 0$, the expansions $V'(t) = t/3 + O(t^2)$ and $j'(t) = t + O(t^2)$ show $R_s(t) \to -1/3$. As $t \to \infty$, uniformly for $0 \le s \le s_0 < 1$,

$$R_s(t) = \frac{s \int_0^\infty V(v)\, dv}{t} + O_{s_0}(t^{-2}).$$

To see this, split each integral at $t/2$. On the first part expand $(1 - v/t)^{-s} = 1 + sv/t + O_{s_0}(v^2/t^2)$. On the second part use the exponential decay of V', j' and the integrability of $(t-v)^{-s}$, uniform for $s \le s_0$. Now $\int V' = 0$, $\int vV' = -\int V$, and $\int j' = 1$. This proves the expansion. Thus for $s > 0$ the maximum is positive, finite, and attained at some interior t. On compact s-subintervals of $(0, 1)$ the two endpoint bounds are uniform and the quotient is jointly continuous on compact interior rectangles. Consequently C is continuous there.

If $0 < s_1 < s_2 < 1$, the fractional integral semigroup identity makes $R_{s_1}(t)$ a weighted average of $R_{s_2}(v)$, $0 < v < t$, with strictly positive weight $(t-v)^{s_2 - s_1 - 1} L_{s_2} j'(v)$. The identity follows by interchanging integrals and using the beta integral. Evaluate at a maximizer for s_1. Since the averaged quotient is negative near $v = 0$ but $C(s_2) > 0$, strict inequality $C(s_1) < C(s_2)$ follows. When $s \downarrow 0$, the compact-interior limit is $R_0 = -V/j < 0$, and the uniform endpoint estimates give $C(s) \to 0$. When $s \uparrow 1$, $L_s u(t) \to u(t)$ at fixed $t > 0$ for continuous u, by concentration of the normalized power kernel. Therefore $\liminf_{s \uparrow 1} C(s) \ge -V'(t)/j'(t)$ for every t. The latter quotient grows as $t^2/6$ by Lemma 11.2. Hence $C(s) \to \infty$.

Finally Lemma 11.2 proves the two integer degrees are attainable, and the pole orders at zero give their upper bounds. The strictly increasing inverse gives the exact attained degree when $\lambda < 0$, since CM at one power implies CM at every smaller power by multiplication by a CM power. This proves Theorem 11.1.

12 Reproducing the finite checks

Everything needed to typeset this article is in this file. Compile with `pdflatex` twice, or until all cross-references stabilize. After `\end{document}` the source contains five files between matching `%% BEGIN` and `%% END` lines. Copy only the lines between a matching pair of delimiters.

- Run `roy_square_verify.py` with Python 3. It uses exact integer arithmetic. Its finite differences certify the sign of infinitely many coefficients on arithmetic progressions.
- Build `zero_forcing_audit.cpp` using a C++20 compiler and run the resulting binary. It simulates the coloring rule on the 34-vertex example.
- Run `mixture_rational.py` with Python-FLINT. It computes the Bernstein coefficients of eight polynomials over the rationals and checks the sign of each coefficient.

- Run `kanter_verify.py` and `trigamma_degrees.py` with Python 3 and `python-flint` 0.9. The displayed final decimal margins are informational. Each required inequality is tested using Arb balls or exact rationals before a success message is emitted.

The signs on unbounded domains require the analytic tail estimates given in the proofs. The programs for compact sets do not replace those estimates.

A Zero-forcing polynomials of joins of clique unions

Boyer, Brimkov, English, Ferrero, Keller, Kirsch, Phillips and Reinhart, *The zero forcing polynomial of a graph*, Discrete Applied Mathematics 258 (2019), 35–48, doi:10.1016/j.dam.2018.11.033, posed Conjecture 1 in Section 6 of the 2018 preprint. It asserts unimodality of the coefficients of $\mathcal{Z}(G;x)$ for every graph. P. Graham, *A graph whose zero-forcing polynomial is not unimodal*, 1 November 2026, Zenodo 23875153, had already refuted this conjecture with a 73-vertex graph. That construction also uses a join of unions of cliques and the fact that twin vertices cannot both stay white. We give a smaller witness, a formula for uniform clique sizes on each side, and a construction with arbitrarily many deep valleys. The earlier disproof is excluded from the count of new source-conjecture resolutions.

In this polynomial $z(G;k) = [x^k]\mathcal{Z}(G;x)$ counts the initially blue k-element sets for which repeated application of the following rule colors every vertex: a blue vertex with exactly one white neighbor may color that neighbor blue.

Theorem A.1. *Let $a, b, r, s \geq 2$. Form $G = (aK_r) \vee (bK_s)$ by taking the join of two disjoint unions of cliques and write $N = ar + bs$. For the polynomial $F(u) = \sum_{t\geq 0} z(G; N-t)u^t$ one has*

$$\begin{aligned} F(u) = {} & (1+ru)^a + (1+su)^b - 1 \\ & + bsu\big[(1+ru)^a - 1 - (ru)^a\big] \\ & + aru\big[(1+su)^b - 1 - (su)^b\big] - abrsu^2. \end{aligned}$$

The connected 34-vertex graph $(7K_2) \vee (4K_5)$ gives

$$z(G;28) = 13888 \;>\; z(G;29) = 11872 \;<\; z(G;30) = 13785.$$

In particular the conjecture is false even on cographs of diameter two.

Proof. The complement of the initial blue set will be called the white set. Two white vertices in the same constituent clique cannot both eventually be colored: before the first of them changes color every blue vertex adjacent to either is adjacent to both. Thus a forcing set leaves at most one white vertex in each constituent clique. Write i, j for the numbers of initially white vertices on the two sides. If $i, j \geq 2$ no first move is available since all vertices are adjacent to the entire opposite side.

If $j = 0$, each left clique containing a white vertex contains a blue vertex with exactly that one white neighbor. All of the whites can be colored. Interchanging the sides handles $i = 0$. These cases give the first line of the formula. Suppose now that $j = 1$ and $i \geq 1$. If $i < a$, a blue vertex in an entirely blue left clique can color the only white vertex on the right. The previous argument then applies. If $i = a$, every left blue vertex sees two whites and every right blue vertex sees at least $a \geq 2$ whites, so the process cannot start. The contribution for $j = 1$ and $1 \leq i < a$ is the second line. The last line adds the analogous case $i = 1$ and $1 \leq j < b$, removing the duplicated $i = j = 1$ contribution. This exhausts all possible initial colorings. Substitution yields the displayed three exact integers. A join of disjoint unions of cliques is a cograph by the disjoint-union and join characterization. Since both sides have at least two cliques, its diameter is two. □

Two concentrations. The same family can give a valley of arbitrarily large depth. Set $r = 2$, $s = 8$, $a = 2b$ and let $b \to \infty$. The two polynomial terms $P = (1 + 2u)^{2b}$ and $Q = (1 + 8u)^b$ have the same total mass 9^b. Their normalized coefficients are binomial laws with respective means $4b/3$ and $8b/9$, separated by $4b/9$. In the coefficientwise order for nonnegative polynomials, the proof gives

$$P \preceq F, \qquad Q \preceq F, \qquad F \preceq (1 + 8bu)P + (1 + 4bu)Q.$$

At the nearest integer to the midpoint $10b/9$, the coefficients of P, Q and their unit shifts are all $9^b O(e^{-cb})$ for some absolute $c > 0$ by Hoeffding's inequality. In each of the two windows of radius $b/9$ about the means, a coefficient of the corresponding pure polynomial is at least $9^b/(5b)$ when b is large, by the same concentration bound and the number of integer indices in a window. Therefore the valley coefficient divided by the smaller of these two peak coefficients is $O(b^2 e^{-cb})$. This argument concerns the original ordinary color-change rule and the unweighted cardinality enumerator.

Any prescribed number of separated peaks. Fix $h \geq 2$, and choose M tending to infinity through multiples of $\operatorname{lcm}(1, \ldots, h)$. Take the join of h graphs $G_j = a_j K_{r_j}$ with $a_j = M/j$ and $r_j = 3^j - 1$. The same first-move obstruction shows that all white vertices lie in one side, or that they lie in two sides with exactly one white vertex on one of these two sides. In fact if three sides have white vertices, every blue vertex sees at least two whites outside its side. Two sides with at least two whites each give the same obstruction. There is still at most one white per constituent clique. Conversely, the single-side case always forces under this last restriction. For $P_j(u) = (1 + r_j u)^{a_j}$ and $N = \sum_j a_j r_j$, this gives

$$P_j \preceq F \quad (1 \leq j \leq h), \qquad F \preceq (1 + Nu) \sum_{j=1}^{h} P_j.$$

All P_j have mass 3^M. Their means $M(1 - 3^{-j})/j$ are pairwise distinct with minimum spacing $\delta_h M$ for some $\delta_h > 0$. Indeed $(1 - 3^{-j})/j$ is $2/3$ times the mean of the strictly decreasing list $1, 1/3, \ldots, 3^{1-j}$. For each fixed h these binomial distributions put all but $O_h(e^{-c_h M})$ of their mass within distance $\delta_h M/4$ of their respective means, by Hoeffding's inequality. At each intervening midpoint all the coefficients of all P_j and their unit shifts are $O_h(3^M e^{-c'_h M})$. Each mean window contains a coefficient at least $3^M/(2(N + 1))$. Since $N = O_h(M)$, the graph has h separated peaks whose intervening valleys are smaller by arbitrarily large factors. The construction stays within connected cographs of diameter two.

B Finite derivative zero counts and infinite divisibility

T. Hasebe, T. Simon and M. Wang, *Some properties of the free stable distributions*, Ann. Inst. H. Poincaré Probab. Statist. 56 (2020), 296–325, arXiv:1805.01133, Remark 15(b), define BS_n by the requirement that the kth derivative of a smooth density on $(0, \infty)$ have exactly $\min(k, n)$ distinct zeros there for every $k \geq 0$. They conjecture that all such densities are classically infinitely divisible for every $n \geq 2$. The same remark identifies the class with the earlier WBS_{n-1} of T. Simon, *Positive stable densities and the bell-shape*, arXiv:1302.0956. That definition specifies endpoint limits in addition to interior zero counts. The two formulations have different answers. The positive conclusion with those extra conditions was already established by Theorem 1.13 and Corollary 1.4 of M. Kwaśnicki and T. Simon, *Characterisation of the class of bell-shaped functions*, arXiv:1910.07752. We give an explicit counterexample to the separate zero-count definition and a direct Gaussian-variation proof to pinpoint the role of the endpoint.

Theorem B.1. *For every $0 < a \leq 1/10$, the probability density proportional to $e^{-x} x (x + a)^7$ on $x > 0$ belongs to the printed class BS_8 and is not infinitely divisible. Conversely, if $f \in \mathrm{BS}_n$ and*

$f^{(j)}(0+) = 0$ *for* $0 \le j < n$*, then its law is infinitely divisible. The latter conclusion includes the earlier endpoint convention in* WBS_{n-1}*.*

Proof. First consider a real monic polynomial P of degree n with all roots in $(-\infty, 0]$. Write D for differentiation and $T = 1 - D$. The kth derivative of $e^{-x}P(x)$ is $(-1)^k e^{-x} T^k P(x)$. For any real-rooted monic polynomial Q with distinct roots r_i of multiplicities m_i, the roots of TQ comprise r_i with multiplicities $m_i - 1$ and the solutions of $\sum_i m_i/(x - r_i) = 1$. There is one simple solution in each gap (r_i, r_{i+1}) and one to the right of the largest root. Consequently each iteration increases the number of strictly positive roots, counted with multiplicity, by at most one. Once all n roots are positive and simple, they stay positive and simple at every subsequent iteration.

Apply this to $P_a = x(x + a)^7$ with $n = 8$. At $a = 0$, $T^k x^n$ has a zero of multiplicity $n - k$ at zero for $k < n$ and exactly k positive simple roots for $k \le n$, by the same gap argument starting from x^n. By continuity of the coefficients, each distinct positive simple root persists under sufficiently small positive a. There are only finitely many orders $k \le 8$ to check. Together with the upper count and propagation for $k > 8$, this proves the required counts for *all* derivatives. This count extends through $0 < a \le 1/10$. Indeed, for $1 \le k \le 8$ the derivative at zero is the alternating sum with leading positive term and unsigned terms, in descending j,

$$\frac{k!}{(k-j)!}\binom{7}{j-1}a^{8-j}, \qquad k \ge j \ge 1.$$

The ratio of each succeeding term to its predecessor is $a(j-1)/[(9-j)(k-j+1)] \le 7a < 1$, so no such derivative has a root at zero. In this parameter interval the positive root counts cannot change: all roots remain real and bounded on compact parameter intervals, and none can cross zero. Positive roots stay simple by induction in k and the gap rule. In particular the example vanishes at the left endpoint although $f'(0+) > 0$.

The unnormalized Laplace transform at s is $(1+s)^{-9}Q(a(1+s))$, by integrating the polynomial termwise, where

$$Q(y) = y^7 + 14y^6 + 126y^5 + 840y^4 + 4200y^3 + 15120y^2 + 35280y + 40320.$$

This polynomial has a root z with $\Re z > 0$. Indeed its fifth Hurwitz determinant, in the descending-coefficient convention, is

$$\det\begin{pmatrix} 14 & 840 & 15120 & 40320 & 0 \\ 1 & 126 & 4200 & 35280 & 0 \\ 0 & 14 & 840 & 15120 & 40320 \\ 0 & 1 & 126 & 4200 & 35280 \\ 0 & 0 & 14 & 840 & 15120 \end{pmatrix} = -423901900800.$$

Were all roots in the closed left half-plane, $Q(y + u)$ for every $u > 0$ would be strictly Hurwitz stable, so all its leading Hurwitz determinants would be positive. Continuity as $u \downarrow 0$ contradicts the display. In fact a root satisfies $\Re z > 1/10$: the same fifth minor for $Q(y + 1/10)$ is $-2697189294631175490229 3/61035156250 < 0$ by binomial expansion. For $0 < a < \Re z$, the Laplace transform therefore has the zero $s = z/a - 1$ in the open right half-plane.

For completeness no nonnegative infinitely divisible law can have such a zero. In every m-fold equal convolution decomposition, the summands are nonnegative almost surely, since the essential infimum of the sum of independent identically distributed variables is m times the common essential infimum. Their Laplace transforms are holomorphic on the open right half-plane. The multiplicity of any zero of the transform of the full law must thus be divisible by every positive integer m, an impossibility. This proves the first assertion.

Gaussian smoothing and endpoint derivatives. For the second assertion, extend the density by zero to $\mathbb{R}$. Every derivative tends to zero at $+\infty$. In fact f is eventually monotone and integrable there. Inductively $f^{(j)}$ is eventually monotone and any nonzero or infinite limit would contradict the preceding derivative's vanishing limit. Because $f^{(j)}$ has constant sign on finitely many intervals, the endpoint conditions imply integrability of $f^{(j)}$ on the half-line for $1 \le j \le n$. This follows by integrating on each such interval and using the endpoint limits of $f^{(j-1)}$. The distributional jth derivative of the extension is the zero extension of $f^{(j)}$ for $j \le n$, with no boundary atoms, by integration by parts.

Let p be any centered Gaussian density of positive variance. For $k \le n$ the Gaussian variation-diminishing property gives at most k sign changes of $(p * f)^{(k)}$, by applying it to $f^{(k)}$ extended by zero. For $k = n + d$, $d > 0$, write $h = f^{(n)}\mathbf{1}_{(0,\infty)}$ (for $n = 0$ take $h = f$). It is integrable, so $h_\epsilon = h\mathbf{1}_{[\epsilon,\infty)}$ converges in L^1 to h. Its distributional dth derivative consists of $f^{(k)}$ on (ϵ, ∞) together with a linear combination of point-mass derivatives at ϵ of orders at most $d - 1$. The regular part is an integrable signed measure there since $f^{(k-1)}$ tends to zero and $f^{(k)}$ has finitely many zeros. The point-supported part can be approximated by a signed measure on the d points $\epsilon - j\delta$, $1 \le j \le d$, as $\delta \downarrow 0$. Indeed, match its action on polynomials through degree $d - 1$ by solving a Vandermonde system. The weights are $O(\delta^{1-d})$ and the Taylor remainder then gives convergence against smooth tests, including each translate of the Gaussian.

The approximating measures including the regular part have at most $d + n = k$ sign changes. There are at most $d - 1$ among the added atoms, at most one at the join, and at most n on the right. Gaussian convolution cannot increase the number of sign changes. This classical variation-diminishing theorem follows from the total positivity of the Gaussian kernel; see I. J. Schoenberg, *On Pólya frequency functions. I*, J. Analyse Math. 1 (1951), 331–374. Passing first to the pointwise limit $\delta \downarrow 0$ and then to $\epsilon \downarrow 0$ shows that $p^{(d)} * h = (p * f)^{(k)}$ also has at most k sign changes. For the second limit use boundedness of $p^{(d)}$ and the L^1 convergence. A strict alternation at finitely many sample points persists under pointwise convergence, which justifies both upper bounds at the limit.

All derivatives of $p * f$ tend to zero at both infinities by integrability of f, and $p * f$ is positive. Successive mean-value arguments give at least k sign changes of its kth derivative: each sequence of alternating nonzero values for the preceding derivative gives one more such alternation for the derivative by taking differences between successive values and using zero limits at both ends. Thus all Gaussian smoothings are strictly bell-shaped in the sense of M. Kwaśnicki and T. Simon, *Characterisation of the class of bell-shaped functions*, arXiv:1910.07752. Their Corollary 1.4 implies infinite divisibility. Equivalently the zero-extended original probability is weakly bell-shaped and that corollary applies directly to it. □

The alternative proof of the known endpoint implication uses an established characterisation after explicitly controlling the point-mass derivatives of a truncated half-line density. The negative example addresses only the definition in Remark 15(b) of the 2020 source. It is compatible with the authors' separate gamma-mixture prediction, since the example divided by x^8 is proportional to $e^{-x}(1 + a/x)^7$, a completely monotone function.

C The gamma-mixture question for finite bell shape

Hasebe, Simon and Wang, Remark 15(b) cited in Section B, also predict that every BS_n density is a scale mixture of gamma densities of shape $n+1$. This statement concerns a different property of the law. The next theorem distinguishes the two notions of zero counts in this question. To be a gamma mixture of this shape a probability density f must and need only have $x^{-n}f(x)$ completely monotone. Indeed each positive-rate gamma density has the form $v^{n+1}x^n e^{-vx}/n!$, and conversely normalization of a positive Laplace representation gives a probability measure on those rates by Tonelli's theorem. For the stronger endpoint convention, Kwaśnicki–Simon, Theorem 1.13,

already give a convolution of n exponentials and a completely monotone density. Conditional on the mixing rate of the latter, its random variable is a sum of $n+1$ independent exponentials of possibly distinct rates. Writing unit exponentials as their sum times their independent Dirichlet proportions exhibits this as a gamma mixture of shape $n+1$. The result below allows all lower endpoint jets to be nonzero.

Theorem C.1. *If for every $j \geq 0$ the derivative $f^{(j)}$ has exactly $\min(j, n)$ distinct zeros in $(0, \infty)$ and changes sign at each of those zeros, then f is a gamma mixture of shape $n+1$. Under the published zero-count definition without this sign-change condition there is a counterexample already for $n = 3$.*

Proof. For the counterexample, take f proportional to $e^{-x}x((x-1)^2+6)$ on $(0,\infty)$. This is positive and has $f(0+) = 0$. Ignoring the fixed positive normalization, its first two derivatives are

$$f' = -e^{-x}(x-1)((x-2)^2+3), \qquad f'' = e^{-x}(x-2)(x-3)^2.$$

The gap argument for $1-D$ in Theorem B.1 gives three simple positive zeros for f''' and every subsequent derivative. Thus $f \in \mathrm{BS}_3$ in the printed sense. But its quotient by x^3 is proportional to $e^{-x}(1-2/x+7/x^2)$. Its inverse Laplace measure, up to that constant, is

$$\delta_1 + \mathbf{1}_{v>1}\big[-2+7(v-1)\big]\,dv.$$

This has negative mass on intervals immediately to the right of one. Uniqueness of Laplace transforms rules out a positive representing measure, so there is no gamma mixture.

For the positive assertion the tail sign of $f^{(j)}$ is $(-1)^j$. Indeed all derivatives tend to zero at infinity as in the preceding proof. Since $f > 0$ and each derivative is eventually monotone and nonzero, induction gives the alternating signs there. For $j \leq n$ the j sign changes force $f^{(j)} > 0$ near zero. In particular $f^{(r)}(0+)$ exists, is finite and is nonnegative for $0 \leq r < n$: both this derivative and the following derivative are positive near zero.

Fix a derivative order k and set $m = n+k+2$. Define the signed measure μ by pushing forward $(-1)^m t^{m-1} f^{(m)}(t)\,dt/(m-1)!$ under $s = m/t$. For $0 \leq r < n$ it is absolutely integrable against s^r and

$$\int s^r\,d\mu(s) = (-1)^r \frac{m^r(m-1-r)!}{(m-1)!}\, f^{(r)}(0+).$$

These formulas follow by successive integration by parts. Here are endpoint details. If a derivative $f^{(r)}$ has finite limit at an endpoint, then $t^j f^{(r+j)}(t) \to 0$ there for every $j \geq 1$. For $j = 1$ use eventual monotonicity of $f^{(r+1)}$ and secants on $[t/2, t]$ and $[t, 2t]$. Iterating these inequalities with the previous weighted limits proves the assertion for every j. These bounds hold at infinity and at zero for the r in question. Applying integration by parts first on compact intervals proves the formula at the endpoints. Since $f^{(m)}$ has finitely many zeros, convergence of these improper moments entails absolute convergence. For $n = 0$ the assertion follows immediately from the alternating tail signs on the entire half-line; hence assume $n \geq 1$. Taylor's formula with integral remainder at a right endpoint tending to infinity gives the exact identity

$$f(x) = \int (1 - xs/m)_+^{m-1}\,d\mu(s).$$

The Taylor polynomial at the right endpoint vanishes in that limit by the same secant estimates.

For $x > 0$ define

$$H(s) = (-\partial_x)^k \left[x^{-n}(1-xs/m)_+^{m-1}\right].$$

For $0 \leq r \leq n$ one has $(-1)^r H^{(r)}(s) \geq 0$. In fact, applying $(-\partial_s)^r$ first leaves a positive constant times $x^{r-n}(1-xs/m)_+^{m-1-r}$. The first factor has alternating derivatives of all orders, and the second has alternating derivatives in x through order k. Since $m = n+k+2$, these statements

and the differentiations are valid across the cutoff $xs = m$. Differentiating under the measure integral is legitimate on a neighborhood of each $x > 0$: all nonzero terms have s in a common compact interval.

The signs of μ as s increases start positive and alternate at n points $0 < b_1 < \cdots < b_n$. Interpolate H at those points by the polynomial P of degree at most $n-1$. Its Newton divided difference of order j has sign $(-1)^j$ or is zero, by the derivative bound. Each Newton basis product of j factors $s - b_i$ has coefficient of degree r of sign $(-1)^{j-r}$. It follows that $(-1)^r[s^r]P \geq 0$ at every degree r. The moment formulas imply $\int P\,d\mu \geq 0$. The interpolation remainder has sign $(-1)^n \prod_i (s - b_i)$ or is zero, by the mean-value formula for the divided difference of order n. This is precisely the sign of μ. Hence $\int (H - P)\,d\mu \geq 0$ as well. We have shown $(-\partial_x)^k(x^{-n}f(x)) = \int H\,d\mu \geq 0$, which proves the gamma-mixture statement for every k. □

More generally the interpolation proof works whenever the first n endpoint jets are finite and nonnegative and $f^{(m)}$ changes sign at most n times for each m; one uses the actual number $\ell \leq n$ of sign changes as the interpolation order, integrating H directly if $\ell = 0$. The estimates at the endpoints in this argument require the eventual monotonicity of each derivative, which already follows from the assumed finite sign changes of the following derivative. Thus no equality of the first n endpoint jets to zero is required. The source's shape-$n+1$ mixture assertion and its infinite-divisibility assertion are logically distinct: gamma mixtures of arbitrary shape larger than two need not be infinitely divisible, as the authors point out.

D The principal-eigenfunction inflection question of Li, Morgan and Siegrist

Y. Li, R. Morgan and M. Siegrist, *Concavity structure of fractional differential equations and zeros of Mittag–Leffler functions*, J. Anal. (2026), doi:10.1007/s41478-026-01192-9, specify an eigenfunction prediction immediately after their Conjecture 6.1. For $1 < a < 2$ let λ_a be the least positive z such that $E_{a,a}(-z) = 0$, where $E_{a,b}(z) = \sum_{j\geq 0} z^j/\Gamma(aj + b)$. They predict that $E_{a,a-2}(-\lambda_a x^a)$ has exactly one zero in $0 < x < 1$. Their Theorem 6.2 gives existence. We prove uniqueness and simplicity. The same paper's numbered conjecture for arbitrary nonnegative forcing has a different answer. A public counterexample to the latter states separately that the eigenfunction question is open.

Theorem D.1. *Let $1 < a < 2$, let $g(t) = t^{a-1}E_{a,a}(-t^a)$, and put $b = \sin(\pi/a)$. For every integer $k \geq 0$ and every interval $I \subset (0, \infty)$ of length at most π/b, the derivative $g^{(k)}$ has at most two zeros in I, counting multiplicities. The principal eigenfunction $x^{a-1}E_{a,a}(-\lambda_a x^a)$ has exactly one inflection point on $(0, 1)$, and the corresponding zero of $E_{a,a-2}(-\lambda_a x^a)$ is simple.*

Proof. Write $\theta = \pi/a$ and $c = -\cos\theta > 0$. The transform of g is $1/(1 + s^a)$, by its series first on the real axis $s > 1$. Deform the inversion contour about the principal negative real cut. The simple poles $e^{\pm i\theta}$ have residues $-e^{\pm i\theta}/a$. For each $t > 0$ this gives

$$g(t) = -\frac{2}{a}e^{-ct}\cos(bt + \theta) - K(t), \qquad K(t) = \frac{-\sin(\pi a)}{\pi}\int_0^\infty \frac{e^{-rt}r^a}{1 + 2r^a\cos(\pi a) + r^{2a}}\,dr.$$

Indeed the cut is minus the imaginary part of the upper boundary value of the transform, divided by π. The large arc vanishes since $a > 1$, and the small arc at zero vanishes by boundedness. For every k, $K_k = (-1)^k K^{(k)}$ is the strictly positive Laplace transform of the displayed density multiplied by r^k. Its logarithm is convex by Cauchy–Schwarz or the nonnegative variance formula.

After multiplying $g^{(k)}$ by $(-1)^k$ the oscillatory term is $s_k(t) = (-1)^{k+1}(2/a)e^{-ct}\cos(bt + (k + 1)\theta)$, and the other term is $-K_k(t)$. Zeros occur only where $s_k > 0$. Two consecutive such open

lobes are separated by a gap of length π/b, so I can contain zeros from at most one lobe. Within it,

$$(\log(s_k/K_k))'' = -b^2 \sec^2(bt + (k+1)\theta) - (\log K_k)'' < 0.$$

Hence there are at most two zeros counted with multiplicity: the derivative is strictly decreasing, and a stationary zero has nonzero second derivative. The zeros and multiplicities of $s_k - K_k$ agree with those of this log ratio.

Let $T = \lambda_a^{1/a}$. Near zero $g(t) \sim t^{a-1}/\Gamma(a) > 0$. At $L = (3\pi/2 - \theta)/b$ the cosine term vanishes, so $g(L) = -K(L) < 0$ and $T < L < \pi/b$. On $0 < x < 1$ set $u(x) = T^{1-a} g(Tx) > 0$. It is the principal eigenfunction of the source, with homogeneous endpoints, $\mathcal{D}_{0+}^a u = -\lambda_a u$. Its twice differentiated series gives $u''(x) = x^{a-3} E_{a,a-2}(-\lambda_a x^a)$. Theorem 5.2 of Li–Morgan–Siegrist applied to $F = -u$ and $f = \lambda_a u$ gives $u'' < 0$ near zero and $u'' > 0$ near one (their proof gives strict curvature). Its weighted derivative hypothesis holds since $u'(x) = O(x^{a-2})$ and $(3-a)/(2-a) + a - 2 > 0$. The same signs follow from analyticity and the stated strict concavity on neighborhoods, together with the two-zero bound. The bound for $k = 2$ permits at most two zeros counting multiplicity. Opposite endpoint signs require odd total multiplicity, hence exactly one simple zero. The positive factor x^{a-3} preserves it. □

The smallest-modulus question. The discussion before the proof of the source's Theorem 6.3 asks whether the first positive zero of $E_{a,a-2}(-z)$ is a zero of smallest modulus. In fact the unique zero of smallest modulus is strictly negative. In $P(z) = E_{a,a-2}(z)$ the constant term $1/\Gamma(a-2)$ is negative and every coefficient of positive degree is strictly positive. Thus P has a unique positive zero r. The triangle inequality excludes all zeros in $|z| < r$. Equality on $|z| = r$ forces the degree-one and degree-two monomials to have the same phase, giving $z = r$. Replacing z by $-z$ proves the assertion. This last observation only uses elementary coefficient signs.

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